Lecture 1

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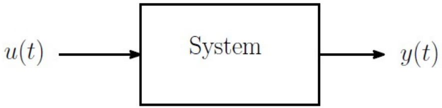
E1 248
Sliding mode control and its application

Prerequisite: Transfer function, Laplace transform, Solution of differential equations

Topics covered:

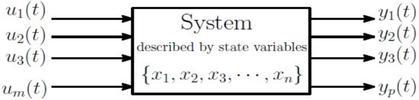
- What is state?
- Why do we need state variables?
- State space model for RLC circuit.
- Invariance of the transfer function for different representations.
- Different state space realization

- **Plant:** A plant may be a piece of equipment, perhaps just a set of machine parts functioning together, the purpose of which is to perform a particular operation.
- **System:** A system is a combination of components that act together and perform a certain objective. The concept of the system can be applied to abstract, dynamic phenomena.
- A system can be physical, biological, economic, etc.
- The mathematical model of a system is linear, if it obeys the principle of superposition and homogeneity.
- **Linear system:** A mathematical model is linear if the differential equation describing it has coefficients, which are either functions only of the independent variable or are constants.
- **Linear time-varying:** If coefficients of the describing differential equations are functions of time (the independent variables), then the mathematical model is linear time-varying.
- **Linear time-invariant:** On the other hand, if the coefficients of the describing differential equations are constants, the model is linear time-invariant.



Definition of state:

The state of a system is the set of variables, which, if known at time t_0 together with input $u(t)$ for $t \geq t_0$, the system behavior for all $t \geq t_0$ can be determined.



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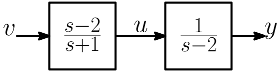
Why we need state variables?

1. This approach can be applied to linear or nonlinear, time-variant or time-invariant systems.
2. Can handle SISO and MIMO systems with equal ease.
3. The internal behavior of the systems is complicated, and it can not be described by a transfer function.
4. The chosen state variables need not represent physical or measurable variables.
5. Matrix transformations can be used to convert from one state variable representation to the other.

Consider the following block diagram:



The transfer function $G(s) = \dfrac{Y(s)}{V(s)} = \dfrac{s-2}{s+1} . \dfrac{1}{s-2} = \dfrac{1}{s+1}$.

- First calculate the step response:
$$v(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \implies V(s) = \frac{1}{s} \implies Y(s) = \frac{1}{s(s+1)}$$
- The output response is solved as $y(t) = \mathcal{L}^{-1}\left[\dfrac{1}{s} - \dfrac{1}{s+1}\right] = 1 - e^{-t}, \ t \geq 0.$
- Is this calculation of $y(t)$ represent true output of the system?
- It can be seen from the actual implementation of this system (on an analog computer) that the output y saturates or the system burns out.
- From the first transfer function, we get
$$\dot{u} = -u + \dot{v} - 2v \implies \dot{u} - \dot{v} = -(u - v) - 3v.$$
- From the second transfer function, we get
$$\dot{y} = 2y + u.$$

- By defining state variables $x_1 = u - v$ and $x_2 = y$, we get the following differential equations

$$\begin{aligned} \dot{x}_1 &= -x_1 - 3v \qquad & x_1(0) = x_{10} \\ \dot{x}_2 &= x_1 + 2x_2 + v \quad & x_2(0) = x_{20}. \end{aligned}$$

- Solve the above differential equation using Laplace transforms:

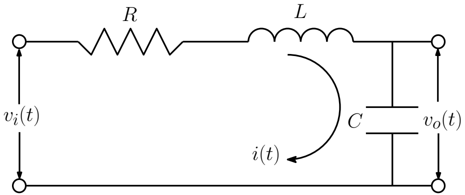
$$\begin{aligned} sX_1(s) - x_1(0) &= -X_1(s) - 3V(s) \\ sX_2(s) - x_2(0) &= X_1(s) + 2X_2(s) + V(s) \end{aligned}$$

- Calculate $Y(s) = X_2(s)$ using above equations as

$$Y(s) = \frac{x_2(0)}{s-2} + \frac{x_1(0)}{(s-2)(s+1)} + \frac{v(s)}{s+1}$$

- This gives $y(t) = e^{2t}x_{20} + \frac{1}{3}(e^{2t} - e^{-t})x_{10} + e^{-t}v(t)$.
- For non-zero initial conditions, the response of the system deviates as $t \to \infty$.

State space model for RLC cricuit



- From Kirchoff's Laws:

$$L\frac{di}{dt} + Ri + v_o(t) = v_i(t)$$
$$\frac{dv_o}{dt} = \frac{1}{C}i(t)$$

- Using the above equations, one can write

$$LC\frac{d^2v_o}{dt^2} + RC\frac{dv_o}{dt} + v_o(t) = v_i(t).$$

- **Step 1:** Express the highest derivative of $v_o(t)$ in terms of all other quantities

$$\frac{d^2 v_o}{dt^2} = -\frac{R}{L}\frac{dv_o}{dt} - \frac{1}{LC}v_o(t) + \frac{1}{LC}v_i(t)$$

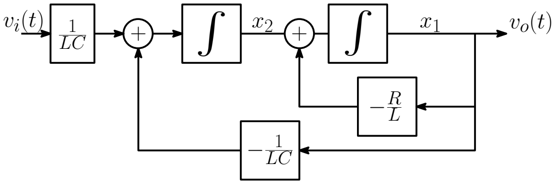
- **Step 2:** Integrate both sides $n = 2$ times

$$\iint \ddot{v}_o(t) = -\frac{R}{L}\iint \dot{v}_o(t) - \frac{1}{LC}\iint v_o(t) + \frac{1}{LC}\iint v_i(t)$$

- **Step 3:** Further reduce to represent above equation only in v_o and v_i

$$v_o(t) = -\frac{R}{L}\int v_o(t) - \frac{1}{LC}\iint v_o(t) + \frac{1}{LC}\iint v_i(t) \tag{1}$$

- **Step 4:** Draw $n = 2$ integrators. Put output (v_o) at the right end and input (v_i) at the left end.
- **Step 5:** Draw above block diagram according to (1).



- Label the output of the integrators as x_1, x_2.
- x_1, x_2 state variables and they are related to v_o and v_i through a set of first-order differential equations.

State equations

$$\dot{x}_1(t) = -\frac{R}{L}x_1(t) + x_2(t)$$
$$\dot{x}_2(t) = -\frac{1}{LC}x_1(t) + \frac{1}{LC}v_i(t)$$
$$y(t) = v_o(t) = x_1(t)$$

State space model

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-\frac{R}{L} & 1\\ -\frac{1}{LC} & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{LC}\end{bmatrix} v_i$$
$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

- A differential equation of order n can be represented as n first-order differential equations.

- It is possible to write the same differential equation in different state space representations.
- Let $x_1 = v_o$ and $x_2 = \frac{dv_o}{dt}$.
- Calculate $\dot{x}_2$ from $\frac{d^2 v_o}{dt^2} = -\frac{R}{L}\frac{dv_o}{dt} - \frac{1}{LC}v_o(t) + \frac{1}{LC}v_i(t)$.

State equations

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -\frac{1}{LC}x_1(t) - \frac{R}{L}x_2(t) + \frac{1}{LC}v_i(t)$$

$$y(t) = v_o(t) = x_1(t)$$

State space model

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{1}{LC} & -\frac{R}{L}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{LC}\end{bmatrix}v_i$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

- Standard choice of states:
  For a mechanical system: Displacement $x(t)$ and Velocity $v(t)$.
  For an electrical system: Inductor current $i(t)$ and Capacitor voltage $v(t)$.

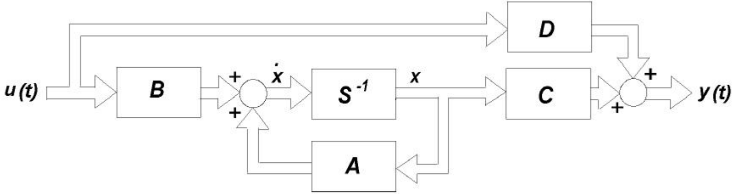


Figure: Block diagram representation of state model

The general form of state equation:

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{2a}$$

$$y(t) = Cx(t) + Du(t) \tag{2b}$$

- x is n-dim. state vector
- y is p-dim. output vector
- u is m-dim. input vector

- A is $n \times n$ system matrix
- B is $n \times m$ control matrix
- C is $p \times n$ output matrix
- D is $p \times m$ feedforward matrix

In some cases, there is no direct connection between input and output, so D will become zero.

Invariance of transfer function

Laplace transform of (2) gives

$$\begin{aligned} sX(s) - x(0) &= AX(s) + BU(s) \\ Y(s) &= CX(s) \\ \implies (sI - A)X(s) &= x(0) + BU(s) \text{ for } x(0) = 0 \\ \implies X(s) &= (sI - A)^{-1}BU(s) \\ \implies Y(s) &= C(sI - A)^{-1}BU(s) \\ \implies G(s) &= \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B. \end{aligned}$$

Invariance of transfer function

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$$
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sX(s) - x(0) &= AX(s) + BU(s) \\
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\implies X(s) &= (sI - A)^{-1}BU(s) \\
\implies Y(s) &= C(sI - A)^{-1}BU(s) \\
\implies G(s) &= \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B.
\end{aligned}
$$

Let there exists a transformation matrix T such that $x = T\hat{x}$, so

$$
\begin{aligned}
\dot{\hat{x}} &= T^{-1}\dot{x} = T^{-1}Ax + T^{-1}Bu \\
&= T^{-1}AT\hat{x} + T^{-1}Bu \\
\dot{\hat{x}}(t) &= \hat{A}\hat{x}(t) + \hat{B}u(t) \\
y(t) &= CT\hat{x} = \hat{C}\hat{x}(t).
\end{aligned}
$$

- For the RLC circuit example,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_o(t) \\ \dot{v}_o(t) \end{bmatrix}, \qquad \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} v_o(t) \\ \hat{x}_2 \end{bmatrix}, \qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{R}{L} & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$$

- Transfer function of new representation:

$$\begin{aligned} \hat{G}(s) &= \hat{C}(sI - \hat{A})^{-1}\hat{B} \\ &= CT\left(sI - T^{-1}AT\right)^{-1} T^{-1}B \\ &= CT\left(sT^{-1}T - T^{-1}AT\right)^{-1} T^{-1}B \\ &= CT\left(T^{-1}(sI - A)T\right)^{-1} T^{-1}B \\ &= CT\left(T^{-1}(sI - A)^{-1}T\right) T^{-1}B \\ &= C(sI - A)^{-1}B = G(s). \end{aligned}$$

- Hence, the transfer function does not change for different representations.
- The new state space representation still represents the same system.

Input-Output Relations-Transfer Functions

- Let the state space description of a system be

$$\dot{X} = AX + BU$$
$$Y = CX$$

  The above equation describes the internal behavior of the system.

- The transfer function of this system is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B$$

  The transfer function describes the external behavior of the system, which is the input-output behavior.

First Companion Form

- Consider the following transfer function

$$\frac{Y(s)}{U(s)} = \frac{1}{s^k + a_1 s^{k-1} + \ldots + a_k}$$

which can be written as

$$\left(s^k + a_1 s^{k-1} + \ldots + a_k\right) Y(s) = U(s) \tag{3}$$

- Differential equation corresponding to (3) is

$$\begin{aligned} D^k y + a_1 D^{k-1} y + \ldots + a_k y = u \\ \Rightarrow D^k y = -a_1 D^{k-1} y - a_2 D^{k-2} y \ldots - a_k y + u \end{aligned} \tag{4}$$

Assume states x_1, x_2, $\cdots$ such that

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ . \quad . &\quad . \\ . \quad . &\quad . \\ \dot{x}_{k-1} &= x_k \end{aligned}$$

Now from (4), following differential can be obtained

$$\dot{x}_k = -a_k x_1 - a_{k-1} x_2 - ... - a_1 x_k + u$$

The matrices corresponding to the system

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdot & \cdot & 0 \\ 0 & 0 & 1 & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -a_k & -a_{k-1} & \cdot & \cdot & \cdot & -a_1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 1 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

- Consider the more general transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^k + b_1 s^{k-1} + ... + b_k}{s^k + a_1 s^{k-1} + ... + a_k} \tag{5}$$

- The development is aided by the introduction of an intermediate variable $Z(s)$ as follows

$$\frac{Y(s)}{U(s)} = \frac{Y(s)}{Z(s)} \cdot \frac{Z(s)}{U(s)} = \frac{b_0 s^k + b_1 s^{k-1} + ... + b_k}{s^k + a_1 s^{k-1} + ... + a_k}$$

- Identify the first factor with the numerator and the second factor with the denominator, i.e.

$$\frac{Y(s)}{Z(s)} = b_0 s^k + b_1 s^{k-1} + ... + b_k \tag{6}$$

and

$$\frac{Z(s)}{U(s)} = \frac{1}{s^k + a_1 s^{k-1} + ... + a_k} \tag{7}$$

Realization of the transfer function from u to z has already been developed.

- From (6)

$$Y(S) = \left(b_0 s^k + b_1 s^{k-1} + ... + b_k\right) Z(S)$$

$$\Rightarrow \; y = b_0 D^k z + b_1 D^{k-1} z + ... + b_k z$$

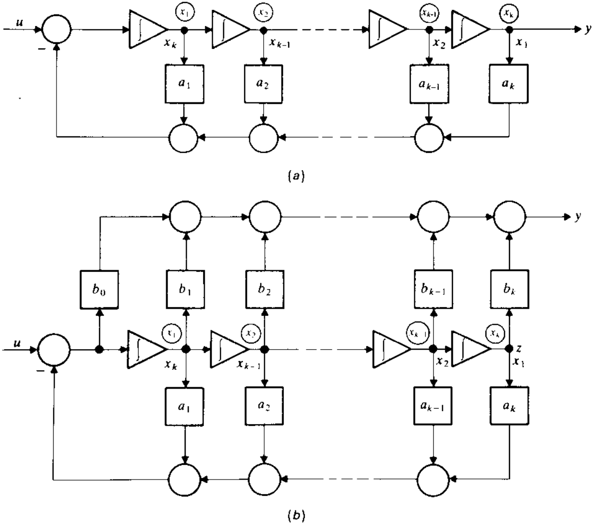

Figure: State-space realization of transfer functions in first companion form for (a) $H(s) = \frac{1}{s^k + a_1 s^{k-1} + \ldots + a_k}$ (b) $H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \ldots + b_k}{s^k + a_1 s^{k-1} + \ldots + a_k}$

- The inputs to the integrators in the chain are the k successive derivatives of z, as shown in the above Figure.
- The state equations are the same as before. The output equation is found by careful examination of the block diagram.
- Note that there are two paths from the output of each integrator to the system output. One path upward through the box labeled b_i and a second path down through the box labeled a_i.
- As a consequence, the output equation is written as

$$y = (b_k - a_k b_0)\, x_1 + (b_{k-1} - a_{k-1} b_0)\, x_2 + \ldots + (b_1 - a_1 b_0)\, x_k + b_0 u$$

Hence

$$C = \begin{bmatrix} b_k - a_k b_0 & b_{k-1} - a_{k-1} b_0 & \cdot & \cdot & \cdot & b_1 - a_1 b_0 \end{bmatrix}$$
$$D = [b_0]$$

- If the direct path through b_0 is absent then the matrix D is zero and the C matrix contains only the b_i coefficients.
- In the first canonical form realization, the input is connected directly to the first integrator in the chain and the output is a linear combination of the outputs of the integrator.

A variant of the structure of the previous Figure in which the output is taken directly from the last integrator, but the input is connected to all the integrators is shown in the figure below.

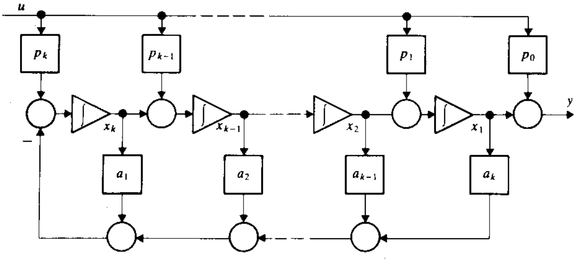

Figure: Alternative first companion form of realization of transfer function $H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \ldots + b_k}{s^k + a_1 s^{k-1} + \ldots + a_k}$

- The feed-forward gains $p_1, p_2, ..., p_k$ are in general not equal to coefficients $b_1, b_2, ..., b_k$ of transfer function, but must be obtained by solution of a set of linear algebraic equations, which may be derived as follows.

$$\begin{aligned}
\dot{x}_1 &= x_2 + p_1 u \\
\dot{x}_2 &= x_3 + p_2 u \\
. \quad . \quad &. \\
. \quad . \quad &. \\
\dot{x}_{k-1} &= x_k + p_{k-1} u \\
\dot{x}_k &= -a_1 x_k - ... - a_k x_1 + p_k u
\end{aligned} \tag{8}$$

and

$$y = x_1 + p_0 u \tag{9}$$

- Differentiating (9) k times and using (8), we obtain

$$\begin{aligned}
Dy &= x_2 + p_1 u + p_0 Du \\
D^2 y &= x_3 + p_2 u + p_1 Du + p_0 D^2 u \\
. \quad . \quad &. \\
. \quad . \quad &. \\
D^{k-1} y &= x_k + p_{k-1} u + p_{k-2} Du + ... + p_1 D^{k-2} u + p_0 D^{k-1} u
\end{aligned}$$

- From (10) and (9) we thus get

$$\begin{aligned} D^k y + a_1 D^{k-1} y + ... + a_{k-1} D y + a_k y = & \left(p_k + a_1 p_{k-1} + ... + a_{k-1} p_1 + a_k p_0\right) u \\ & + \left(p_{k-1} + ... + a_{k-2} p_1 + a_{k-1} p_0\right) D u \\ & + ... + \left(p_1 + a_1 p_0\right) D^{k-1} u + p_0 D^k u \end{aligned} \tag{11}$$

- In order for (11) to represent the original transfer function, it is necessary that

$$\begin{aligned} p_0 &= b_0 \\ p_1 + a_1 p_0 &= b_1 \\ . \quad . & \quad . \\ . \quad . & \quad . \\ p_{k-1} + ... + a_{k-2} p_1 + a_{k-1} p_0 &= b_{k-1} \\ p_k + a_1 p_{k-1} + ... + a_{k-1} p_1 + a_k p_0 &= b_k \end{aligned}$$

which constitute a set of $k+1$ simultaneous equations for $p_0, p_1, ..., p_k$.

The above equations may be arranged in vector-matrix form as

$$\begin{bmatrix} 1 & 0 & \cdot & \cdot & 0 \\ a_1 & 1 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{k-1} & a_{k-2} & \cdot & \cdot & 0 \\ a_k & a_{k-1} & \cdot & \cdot & 1 \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \cdot \\ p_{k-1} \\ p_k \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \cdot \\ b_{k-1} \\ b_k \end{bmatrix} \tag{12}$$

- The $(k+1, k+1)$ matrix in the above equation is known as *TOEPLITZ* matrix. The determinant of the matrix is 1. Hence it is always possible to solve for the p_i, given the numerator coefficients b_i.
- It is worth noting that although the state variables in the original first canonical form and in the alternate canonical form are identified with the outputs of the integrators, they are not the same variables. The matrix A is the same for both but not B and C. Of course, they are related by some linear transformation.

Second Companion Form

- In the first companion form, the coefficients of the denominator of the transfer function appear in one of the rows of A.
- This is another set of companion forms in which the coefficients appear in a column of the A matrix.
- Let us write the original transfer function

$$\left(s^k + a_1 s^{k-1} + \ldots + a_k\right) Y(s) = \left(b_0 s^k + b_1 s^{k-1} + \ldots + b_k\right) U(s)$$

or

$$s^k \left[Y(s) - b_0 U(s)\right] + s^{k-1}\left[a_1 Y(s) - b_1 U(s)\right] + \ldots + \left[a_k Y(s) - b_k U(s)\right] = 0$$

- On dividing by s^k and solving for $Y(s)$, we obtain

$$Y(s) = b_0 U(s) + \frac{1}{s}\left[b_1 U(s) - a_1 Y(s)\right] + \ldots + \frac{1}{s^k}\left[b_k U(s) - a_k Y(s)\right] \quad (13)$$

- Notify that the multiplier $\frac{1}{s^j}$ is the transfer function of a chain of j integrators, immediately leads to the structure as shown in the Figure below.

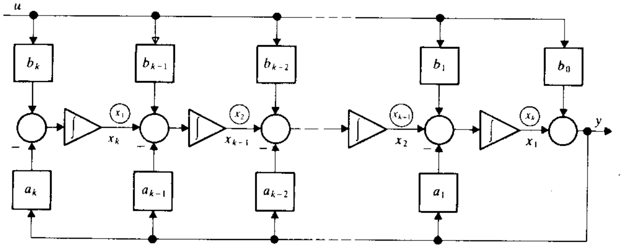

Figure: State-space realization of transfer functions in Second companion form for $H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \ldots + b_k}{s^k + a_1 s^{k-1} + \ldots + a_k}$

- The signal y is fed back to each of the integrators in the chain, and the signal u is fed forward.
- Thus the signal $b_k u - a_k y$ passes through k integrators, as required by (13), the signal $b_{k-1} u - a_{k-1} y$ passes through $k - 1$ integrators and so forth to complete the realization of (13).
- Using the state variables, the differential equations are

$$
\begin{aligned}
\dot{x}_1 &= x_2 - a_1 (x_1 + b_0 u) + b_1 u \\
\dot{x}_2 &= x_3 - a_2 (x_1 + b_0 u) + b_2 u \\
. \quad . &\quad . \\
. \quad . &\quad . \\
\dot{x}_{k-1} &= x_k - a_{k-1} (x_1 + b_0 u) + b_{k-1} u \\
\dot{x}_k &= -a_k (x_1 + b_0 u) + b_k u
\end{aligned}
$$

and the output equation is

$$y = x_1 + b_0 u$$

- Thus, the matrices that describe the state space realization are given by

$$A = \begin{bmatrix} -a_1 & 1 & 0 & \cdot & \cdot & 0 \\ -a_2 & 0 & 1 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -a_{k-1} & 0 & 0 & \cdot & \cdot & 0 \\ -a_k & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \cdot \\ \cdot \\ b_{k-1} - a_{k-1} b_0 \\ b_k - a_k b_0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

$$D = [b_0]$$

- Just as there are two versions of the first companion form, there are also two versions of the second companion form. The second version has the structure as shown in the Figure below.

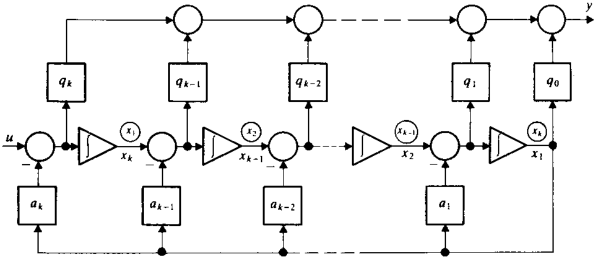

Figure: Alternative second companion form of realization of transfer function $H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \ldots + b_k}{s^k + a_1 s^{k-1} + \ldots + a_k}$

- Here also $q_1, q_2, ..., q_k$ are not the numerator coefficients of the transfer function. They can be evaluated in terms of numerator and denominator coefficients.

Diagonal Form: Partial Fraction Expansion

- Another of the canonical forms of the realization of a transfer function is *diagonal form*, so named because of the nature of the A matrix.
- This canonical form follows directly from the partial fraction expansion of the transfer functions.
- The partial fraction expansion of the transfer function has the form

$$H(s) = b_0 + \frac{r_1}{s - s_1} + \frac{r_2}{s - s_2} + \ldots + \frac{r_k}{s - s_k} \tag{14}$$

- The coefficients $r_i (i = 1, 2, \ldots, k)$ are the residues of the transfer function $H(s) - b_0$ at the corresponding poles.
- In the form of (14), the transfer function consists of a direct path with gain b_0 and k first order transfer functions in parallel.

A block diagram representation of (14) is shown in Figure below as follows.

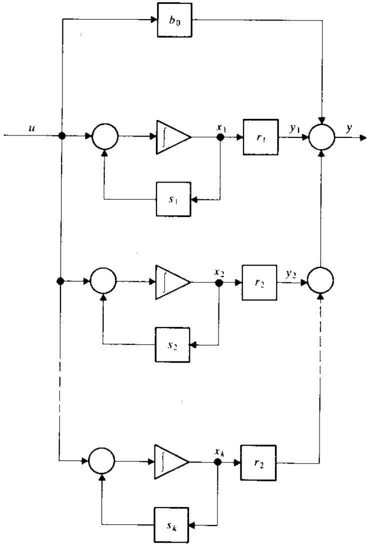

Figure: Diagonal form transfer function with distinct roots

- Identifying the outputs of the integrators with the state variables, results in the following differential equations:

$$
\begin{aligned}
\dot{x}_1 &= s_1 x_1 + u \\
\dot{x}_2 &= s_2 x_2 + u \\
. \quad &. \quad . \\
. \quad &. \quad . \\
\dot{x}_k &= s_k x_k + u
\end{aligned}
$$

and the observation equation

$$y = r_1 x_1 + r_2 x_2 + ... + r_k x_k + b_0 u$$

- Hence, the matrices corresponding to this realization are

$$A = \begin{bmatrix} s_1 & 0 & \cdot & \cdot & 0 \\ 0 & s_2 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & \cdot & s_k \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 1 \\ \cdot \\ \cdot \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} r_1 & r_2 & \cdot & \cdot & r_k \end{bmatrix}$$

$$D = [b_0]$$

Note that A is a diagonal matrix.

- If the system has repeated eigenvalues, then $H(s)$ can be written as

$$H(s) = b_0 + \sum_{i=1}^{r} \frac{C_i}{(s-\lambda_1)^{r-i+1}} + \sum_{i=r+1}^{k} \frac{C_i}{s-\lambda_i}$$

$$\dot{X} = JX + BU$$
$$Y = CX + b_0 U$$

where

$$J = \begin{bmatrix} \lambda_1 & 1 & 0 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_1 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdot & \cdot & \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{r+1} & 0 & \cdot & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{r+2} & \cdot & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \lambda_{r+k} \end{bmatrix}$$

$$B = [0\ 0\ \cdot\ \cdot\ 1\ 1\ 1\ \cdot\ \cdot\ \cdot\ 1]^T$$
$$C = [c_1\ c_2\ \cdot\ \cdot\ c_r\ c_{r+1}\ \cdot\ \cdot\ c_k]$$

This representation is known as the *Jordan Canonical* state model.

Thank You

Lecture 2

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E1 248
Sliding mode control and its application

Overview

1. Solution of the time-invariant state equation

2. Numerical Example

3. Discrete-time systems

Solution of the time-invariant state equation

Before we solve the vector matrix differential equations, let us review the solution of the scalar differential equation

$$\dot{x} = ax$$

In solving this equation, we may assume a solution $x(t)$ of the form

$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

Solution of the time-invariant state equation

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$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

By substituting this assumed solution in the above equation, we obtain

$$b_1 + 2b_2 t + 3b_3 t^2 + \cdots + kb_k t^{k-1} + \cdots = ab_0 + ab_1 t + ab_2 t^2 + \cdots + ab_k t^k + \cdots$$

Equating the coefficient of the equal power of t we obtain

$$
\begin{aligned}
b_1 &= ab_0 \\
b_2 &= \frac{1}{2}ab_1 = \frac{1}{2}a^2 b_0 \\
b_3 &= \frac{1}{3}ab_2 = \frac{1}{6}a^3 b_0 \\
&\vdots \\
b_k &= \frac{1}{k!}a^k b_0
\end{aligned}
$$

Equating the coefficient of the equal power of t we obtain

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&\vdots \\
b_k &= \frac{1}{k!}a^k b_0
\end{aligned}
$$

The value of b_0 is determined by substituting $t = 0$, one can get $b_0 = x(0)$. So the solution can be written as

$$
\begin{aligned}
x(t) &= b_0 + ab_o t + \frac{1}{2}a^2 b_0 t^2 + \cdots \\
&= (1 + at + a^2\frac{t^2}{2!} + \cdots)b_0 \\
&= e^{at}b_0 = e^{at}x(0)
\end{aligned}
$$

We shall now solve the vector matrix differential equation

$$\dot{x} = Ax$$

where $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$

By analogy with the scalar case, we assume that the solution is in the form of a vector power series in t

$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

If the assumed solution is to be the true solution, then

$$b_1 + 2b_2 t + 3b_3 t^2 + \cdots + kb_k t^{k-1} + \cdots = A(b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots)$$

Equating the coefficient of the equal power of t we obtain

$$
\begin{aligned}
b_1 &= Ab_0 \\
b_2 &= \frac{1}{2}Ab_1 = \frac{1}{2}A^2 b_0 \\
b_3 &= \frac{1}{3}Ab_2 = \frac{1}{6}A^3 b_0 \\
&\vdots \\
b_k &= \frac{1}{k!}A^k b_0
\end{aligned}
$$

By substituting $t = 0$ in $x(t)$ one can get $x(0) = b_0$. Thus the solution

$$
\begin{aligned}
x(t) &= (I + At + A^2\frac{t^2}{2!} + A^3\frac{t^3}{3!} \cdots + A^k\frac{t^k}{k!} + \cdots)x(0) \\
&= e^{At}x(0) \neq x(0)e^{At}
\end{aligned}
$$

Write $x(t) = \phi(t)x(0)$ where $\phi(t)$ is an $n \times n$ matrix and is the unique solution of

$$\dot{\phi}(t) = A\phi(t), \qquad \phi(0) = I$$

Note that $x(0) = \phi(0)x(0) = Ix(0)$ and $\dot{x}(t) = \dot{\phi}(t)x(0) = A\phi(t)x(0) = Ax(t)$

So, from the solution of the state equation

$$\phi(t) = e^{At} = \mathscr{L}^{-1}\{(sI - A)^{-1}\}$$
$$\phi^{-1}(t) = (e^{At})^{-1} = e^{-At} = \phi(-t)$$

Properties of state transition matrix for $\dot{x} = Ax$ where $\phi(t) = e^{At}$

1. $\phi(0) = e^{A0} = I$
2. $\phi(t) = e^{At} = (e^{-At})^{-1} = [\phi(-t)]^{-1}$
3. $\phi(t_1 + t_2) = e^{A(t_1+t_2)} = e^{At_1}e^{At_2} = \phi(t_1)\phi(t_2)$

Solution of non-homogeneous state equations

We shall begin by considering the matrix case

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$\dot{x}(t) - Ax(t) = Bu(t)$$

Left multiplication of both sides of this equation by e^{-At} gives

$$e^{-At}(\dot{x}(t) - Ax(t)) = e^{-At}Bu(t)$$
$$\frac{d}{dt}[e^{-At}x(t)] = e^{-At}Bu(t)$$

Integrating this equation from 0 to t gives

$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau)d\tau$$
$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

The first term on the right-hand side is the response to the initial condition, and the second term is the response to the input $u(t)$.

Laplace transform approach for solution of non-homogeneous state equations

The solution of the non-homogeneous state equation can also be obtained by the Laplace transform approach

$$\dot{x}(t) - Ax(t) = Bu(t)$$

The Laplace transform of the above equation yields

$$\begin{aligned} sX(s) - x(0) &= Ax(s) + BU(s) \\ (sI - A)X(s) &= x(0) + BU(s) \\ X(s) &= (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s) \\ x(t) &= \mathscr{L}^{-1}\{(sI - A)^{-1}\}x(0) + \mathscr{L}^{-1}\{(sI - A)^{-1}BU(s)\} \end{aligned}$$

Thus far, we have assumed the initial time is zero. If, however, the initial time is given by t_0 instead of 0, then the solution to the state equation must be modified to

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau$$

Numerical Example

Obtain the time response of the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

for u unit impulse and unit step.

To find state transition matrix $\phi(t) = \mathscr{L}^{-1}\{(sI - A)^{-1}\}$

$$(sI - A) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$= \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

$$\mathscr{L}^{-1}\{(sI-A)^{-1}\} = \mathscr{L}^{-1}\begin{bmatrix} \frac{2}{(s+1)} + \frac{-1}{(s+2)} & \frac{1}{(s+1)} + \frac{-1}{(s+2)} \\ \frac{-2}{(s+1)} + \frac{2}{(s+2)} & \frac{-1}{(s+1)} + \frac{2}{(s+2)} \end{bmatrix}$$

The state transition matrix $\phi(t) = e^{At}$ was obtained earlier

$$\phi(t) = e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

The response to the unit impulse input is then obtained as

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}B\delta(\tau)d\tau.$$

As $\int_{0-\Delta}^{0+\Delta} f(t)\delta(t)dt = f(0)$ we have

$$x(t) = e^{At}(x(0) + B)$$

If the initial condition of state is zero $x(0) = 0$, then impulse response $x(t)$ is

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$$

The response to the unit step input is then obtained as

$$x(t) = e^{At}x(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

If the initial condition of state is zero, or $x(0) = 0$, then $x(t)$ can be simplified to

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

Solution of Linear Time-Varying Systems

- Time-varying system can be solved from state space representation.
- This can be done by changing the transition matrix to $\phi(t, t_0)$ from $\phi(t)$.
- For time-varying systems, the transition matrix depends upon both t and t_0 and not the difference $t - t_0$. Thus, we cannot always set the initial time equal to zero.

Solution of time-varying autonomous state equation

For a Scalar differential equation (1) the solution can be given by (2)

$$\dot{x} = a(t)x(t) \tag{1}$$

$$x(t) = e^{\int_{t_0}^{t} a(\tau)d\tau} x(t_0) \tag{2}$$

The state transition function is given by

$$\phi(t, t_0) = e^{\int_{t_0}^{t} a(\tau)d\tau}$$

The same results, however, do not carry over to the vector matrix differential equation.

Consider the state equation

$$\dot{x} = A(t)x(t)$$

where $x(t)$ is n dimensional vector and $A(t)$ is $n \times n$ matrix whose elements are piecewise continuous function of t in the interval $t_0 \leq t \leq t_1$. The solution of this equation is given by

$$x = \phi(t, t_0)x(t_0)$$

where $\phi(t, t_0)$ is the $n \times n$ nonsingular matrix satisfying the following matrix differential equation

$$\dot{\phi}(t, t_0) = A(t)\phi(t, t_0), \qquad \phi(t_0, t_0) = I$$

The above fact that $x(t) = \phi(t, t_0)x(t_0)$ is the solution of the above equation can be verified easily as given below

$$x(t_0) = \phi(t_0, t_0)x(t_0) = Ix(t_0)$$

and

$$\begin{aligned}\dot{x}(t) &= \frac{d}{dt}[\phi(t, t_0)x(t_0)] \\ &= \dot{\phi}(t, t_0)x(t_0) \\ &= A(t)\phi(t, t_0)x(t_0) = A(t)x(t)\end{aligned}$$

It is important to note that the state transition matrix $\phi(t, t_0)$ is given by a matrix exponential if and only if $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ commute. That is

$$\phi(t, t_0) = e^{\left[\int_{t_0}^{t} A(\tau)d\tau\right]}$$

Note that if $A(t)$ is a constant matrix or diagonal matrix, $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ commute.

If $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ do not commute, then there is no simple way to compute the state transition matrix.

To compute $\phi(t, t_0)$ numerically, we may use the following series expansion for $\phi(t, t_0)$

$$\phi(t, t_0) = I + \int_{t_0}^{t} A(\tau)d\tau + \int_{t_0}^{t} A(\tau_1)\left[\int_{t_0}^{\tau_1} A(\tau_2)d\tau_2\right]d\tau_1 + \cdots$$

It is called the Peano-Baker series.

Properties of the state transition matrix $\phi(t, t_0)$

[1] $\phi(t_2, t_1)\phi(t_1, t_0) = \phi(t_2, t_0)$
To prove this, note that

$$x(t_1) = \phi(t_1, t_0)x(t_0)$$
$$x(t_2) = \phi(t_2, t_0)x(t_0)$$

Also $x(t_2) = \phi(t_2, t_1)x(t_1)$ hence

$$\begin{aligned} x(t_2) &= \phi(t_2, t_1)\phi(t_1, t_0)x(t_0) \\ &= \phi(t_2, t_0)x(t_0) \end{aligned}$$

Thus

$$\phi(t_2, t_1)\phi(t_1, t_0) = \phi(t_2, t_0)$$

[2] $\phi(t_1, t_0) = \phi^{-1}(t_0, t_1)$
To prove this note that

$$\phi(t_1, t_0) = \phi^{-1}(t_2, t_1)\phi(t_2, t_0)$$

If we let $t_2 = t_0$ in this last equation, then

$$\phi(t_1, t_0) = \phi^{-1}(t_0, t_1)\phi(t_0, t_0) = \phi^{-1}(t_0, t_1)$$

Solution of linear time-varying state equations

Let us consider the system

$$\dot{x} = A(t)x(t) + B(t)u(t)$$

The elements of $A(t)$ and $B(t)$ are assumed to be piecewise continuous functions of t in the interval $t_0 \leq t \leq t_1$. In order to obtain the solution, let us put

$$x(t) = \phi(t, t_0)\xi(t)$$

where $\phi(t, t_0)$ is the unique matrix satisfying the following equation

$$\dot{\phi}(t, t_0) = A(t)\phi(t, t_0), \qquad \phi(t_0, t_0) = I$$

Then

$$\begin{aligned}\dot{x}(t) &= \frac{d}{dt}\left[\phi(t,t_0)\xi(t)\right]\\ &= \dot{\phi}(t,t_0)\xi(t) + \phi(t,t_0)\dot{\xi}(t)\\ &= A(t)\phi(t,t_0)\xi(t) + \phi(t,t_0)\dot{\xi}(t)\\ &= A(t)\phi(t,t_0)\xi(t) + B(t)u(t)\end{aligned}$$

Therefore $\phi(t,t_0)\dot{\xi}(t) = B(t)u(t)$ or $\dot{\xi}(t) = \phi^{-1}(t,t_0)B(t)u(t)$. Hence

$$\xi(t) = \xi(t_0) + \int_{t_0}^{t} \phi^{-1}(\tau,t_0)B(\tau)u(\tau)d\tau$$

Since $\xi(t) = \phi^{-1}(t,t_0)x(t) = x(t_0)$. The solution is obtained as

$$x(t) = \phi(t,t_0)x(t_0) + \phi(t,t_0)\int_{t_0}^{t} \phi^{-1}(\tau,t_0)B(\tau)u(\tau)d\tau$$

$$x(t) = \phi(t,t_0)x(t_0) + \int_{t_0}^{t} \phi(t,\tau)B(\tau)u(\tau)d\tau$$

Numerical Example

Compute $\phi(t,0)$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{aligned} \int_0^t A(\tau)d\tau &= \int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} d\tau \\ &= \begin{bmatrix} 0 & t \\ 0 & \frac{t^2}{2} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} \left\{ \int_0^{\tau_1} \begin{bmatrix} 0 & 1 \\ 0 & \tau_2 \end{bmatrix} d\tau_2 \right\} d\tau_1 &= \int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} \begin{bmatrix} 0 & \tau_1 \\ 0 & \frac{{\tau_1}^2}{2} \end{bmatrix} d\tau_1 \\ &= \begin{bmatrix} 0 & \frac{t^3}{6} \\ 0 & \frac{t^4}{8} \end{bmatrix} \end{aligned}$$

State space representation of discrete-time systems

- The state space approach to the analysis of dynamic systems can be extended to the discrete-time case.
- The discrete form of the state space representation is quite analogous to the continuous form.

The most general state space representation of linear discrete-time system is

$$
\begin{aligned}
x(k+1) &= G(k)x(k) + H(k)u(k) \\
y(k) &= C(k)x(k) + D(k)u(k)
\end{aligned}
$$

where $x(k)$ is the state vector, $u(k)$ is the input vector and $y(k)$ is the output vector.

If the linear discrete-time systems are time-invariant then the above equations are modified to

$$
\begin{aligned}
x(k+1) &= Gx(k) + Hu(k) \\
y(k) &= Cx(k) + Du(k)
\end{aligned}
$$

Recursion procedure

In general, difference equations are easier to solve than differential equations because the former can be solved simply by means of a recursion procedure.

Consider the following difference equation -

$$x(k+1) + 0.2x(k) = 2u(k)$$

where $x(0) = 0$ and $u(k) = 1$ for k=0, 1, 2, ... The solution $x(1)$ can be found by recursion as

$$x(1) = -0.2x(0) + 2u(0) = 2$$

Similarly

$$\begin{aligned} x(2) &= -0.2x(1) + 2u(1) = 1.6 \\ x(3) &= -0.2x(2) + 2u(2) = 1.68 \\ x(4) &= -0.2x(3) + 2u(3) = 1.664 \end{aligned}$$

Note that in this method, $x(k+1)$ cannot be computed unless $x(k)$ is known. This procedure is quite simple and convenient for digital computations.

Solution of discrete-time state equations

The state space representation of a discrete LTI system is

$$\begin{aligned} x(k+1) &= Gx(k) + Hu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

The solution of this equation for any $k > 0$ may be obtained directly by recursion as follows.

$$\begin{aligned} x(1) &= Gx(0) + Hu(0) \\ x(2) &= Gx(1) + Hu(1) \\ &= G[Gx(0) + Hu(0)] + Hu(1) \\ &= G^2x(0) + GHu(0) + Hu(1) \\ x(3) &= Gx(2) + Hu(2) \\ &= G[G^2x(0) + GHu(0) + Hu(1)] + Hu(2) \\ &= G^3x(0) + G^2Hu(0) + GHu(1) + Hu(2) \end{aligned}$$

Proceeding similarly, we get

$$\begin{aligned} x(k) &= G^kx(0) + G^{k-1}Hu(0) + \ldots + GHu(k-2) + Hu(k-1) \\ &= G^kx(0) + \sum^{k-1} G^{k-j-1}Hu(j) \end{aligned}$$

Output Equation

- State Transition Matrix: $\Phi(k) = G^k$
- Properties of Φ(k): $\Phi(k+1) = G\Phi(k)$ and $\Phi(0) = I$.

Solution of States

$$x(k) = \Phi(k)x(0) + \sum_{j=0}^{k-1} \Phi(k-j-1)Hu(j) \tag{3}$$

Output Equation

$$\begin{aligned} y(k) &= C\Phi(k)x(0) + C\sum_{j=0}^{k-1} \Phi(k-j-1)Hu(j) + Du(k) \\ &= C\Phi(k)x(0) + C\sum_{j=0}^{k-1} \Phi(j)Hu(k-j-1) + Du(k) \end{aligned}$$

Z-Transform of the state equation

The state space equation of a discrete LTI system is

$$x(k+1) = Gx(k) + Hu(k)$$

Taking Z-transform on both sides of the above equation, we get

$$zX(z) - zX(0) = GX(z) + HU(z) \tag{4}$$

where $X(z) = Z[x(k)]$ and $U(z) = Z[u(k)]$. Then from (4),

$$(zI - G)X(z) = zX(0) + HU(z)$$

Premultiplying the above equation on both sides by $(zI - G)^{-1}$, we get

$$X(z) = (zI - G)^{-1} zX(0) + (zI - G)^{-1} HU(z) \tag{5}$$

Solution & Characteristic Equation

Solution using Z-Transform

Taking the inverse Z-transform of (5), we get

$$x(k) = Z^{-1}(zI - G)^{-1}zX(0) + Z^{-1}(zI - G)^{-1}HU(z)$$

So the state transition matrix

$$G^k = Z^{-1}(zI - G)^{-1}z$$

Characteristic Equation

The characteristic equation of the discrete-time system can be easily noted as

$$|zI - G| = 0$$

Example

Problem

Obtain the state transition matrix of the following discrete-time system

$$x(k+1) = Gx(k) + Hu(k)$$

where

$$G = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then obtain x(k) when u(k)=1 for $k = 0, 1, 2, \ldots$. Assume that the initial condition is given by

$$x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example

Solution

We know that the state transition matrix $\Phi(k)$ is

$$\Phi(k) = G^k = Z^{-1}[(zI - G)^{-1}z]$$

Therefore, we first compute $(zI - G)^{-1}$.

$$(zI - G)^{-1} = \begin{bmatrix} z & -1 \\ 0.16 & z+1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} \frac{z+1}{(z+0.2)(z+0.8)} & \frac{1}{(z+0.2)(z+0.8)} \\ \frac{-0.16}{(z+0.2)(z+0.8)} & \frac{z}{(z+0.2)(z+0.8)} \end{bmatrix}$$

Thus, $\Phi(k)$ is obtained as

$$\Phi(k) = G^k = Z^{-1}[(zI - G)^{-1}z]$$

$$= Z^{-1}\begin{bmatrix} \frac{4}{3}\frac{z}{(z+0.2)} - \frac{1}{3}\frac{z}{(z+0.8)} & \frac{5}{3}\frac{z}{(z+0.2)} - \frac{5}{3}\frac{z}{(z+0.8)} \\ -\frac{0.8}{3}\frac{z}{(z+0.2)} + \frac{0.8}{3}\frac{z}{(z+0.8)} & -\frac{1}{r}3\frac{z}{(z+0.2)} + \frac{4}{3}\frac{z}{(z+0.8)} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4}{3}(-0.2)^k - \frac{1}{3}(-0.8)^k & \frac{5}{3}(-0.2)^k - \frac{5}{3}(-0.8)^k \\ -\frac{0.8}{3}(-0.2)^k + \frac{0.8}{3}(-0.8)^k & -\frac{1}{3}(-0.2)^k + \frac{4}{3}(-0.8)^k \end{bmatrix}$$

Example

Solution Contd...

Next, we compute $x(k)$. The Z-transform of $x(k)$ is given by

$$\begin{aligned} Z[x(k)] = X(z) &= (zI - G)^{-1} zX(0) + (zI - G)^{-1} HU(z) \\ &= (zI - G)^{-1} [zX(0) + HU(z)] \end{aligned}$$

Since $U(z) = \frac{z}{z-1}$, we obtain

$$zX(0) + HU(z) = \begin{bmatrix} z \\ -z \end{bmatrix} + \begin{bmatrix} \frac{z}{z-1} \\ \frac{z}{z-1} \end{bmatrix} = \begin{bmatrix} \frac{z^2}{z-1} \\ \frac{-z^2+2z}{z-1} \end{bmatrix}$$

Hence

$$X(z) = (zI - G)^{-1} [zX(0) + HU(z)] \qquad = \begin{bmatrix} \frac{(z^2+2)z}{(z+0.2)(z+0.8)(z-1)} \\ \frac{(-z+1.84)z^2}{(z+0.2)(z+0.8)(z-1)} \end{bmatrix}$$

Thus,

$$x(k) = Z^{-1}[X(z)] = \begin{bmatrix} -\frac{17}{6}(-0.2)^k + \frac{22}{9}(-0.8)^k + \frac{25}{18} \\ \frac{3.4}{6}(-0.2)^k - \frac{17}{6}(-0.8)^k + \frac{7}{18} \end{bmatrix}$$

THANK YOU

Lecture 3

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E1 248
Sliding mode control and its application

Controllability

- The theory of controllability was proposed in 1960 by R. Kalman.

Definition of Controllability

The linear system with state differential equation

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

is said to be completely controllable if it is possible to construct an unconstrained control signal that will transfer an initial state at any initial time t_0 to any final state $x(t_1) = x_1$ in a finite time interval $t_1 - t_0$.

When we say that the system can be transferred from one state to another, we mean that there exists a piecewise continuous input $u(t), t_0 \leq t \leq t_1$, which brings the system from one state to the other.

A necessary and sufficient condition for controllability in the s plane is that no single pole of the system is cancelled by a zero in all of the elements of the transfer function matrix $[sI - A]^{-1}B$. If such a cancellation occurs, the system cannot be controlled in the direction of the cancelled mode.

Controllability of linear time-invariant Systems

Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(t) \in \mathbb{R}^n.$$

This system is controllable if and only if

$$\mathtt{rank}[B, AB, \cdots, A^{n-1}B] = n.$$

The controllability matrix is defined as

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

Observability

Definition of Observability

Let $y(t; t_0, x_0, u)$ denote the response of the output variable $y(t)$ of the linear differential system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t),$$
$$y(t) = C(t)x(t),$$

to the initial state $x(t_0) = x_0$. Then the system is called completely observable if for all t_1 there exists a t_0 with $-\infty < t_0 < t_1$ such that

$$y(t; t_0, x_0, u) = y(t; t_0, x_0^{'}, u), \qquad t_0 \leq t \leq t_1,$$

for all $u(t)$, $t_0 \leq t \leq t_1$, implies $x_0 = x_0^{'}$.

The system is observable over a time period $[t_0 \ t_1]$ if it is possible to determine uniquely the initial state $x(t_0) = x_0$ from the knowledge of the output $y(t)$ over $[t_0 \ t_1]$.
The complete state of the system is known if the initial state x_0 is known.

Observability of Linear Time-Invariant System

Theorem

Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t),$$
$$y(t) = Cx(t),$$

is completely observable if and only if the row vectors of the Observability matrix

$$Q = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

span the n-dimensional space.

Controllability and Reachability in discrete-time system

Reachability

The system is reachable if the state can be transferred from the zero state at any initial time t_0 to any terminal state $x(t_1) = x_1$ within a finite time $t_1 - t_0$.

Controllability

The system is controllable if the state can be transferred from any state at any initial time t_0 to any terminal state $x(t_1) = 0$ within a finite time $t_1 - t_0$.

In a continuous-time system, controllability and reachability are the same.

Controllability and Reachability

Controllability Need Not Imply Reachability

Consider the system

$$x(k+1) = Ax(k).$$

If A is a nilpotent matrix, then the system states

$$x(k) = A^k x(0)$$

can be driven to zero from any initial state without any input. Such a system is controllable but not reachable.
For one more example of such a system consider

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

then,

$$\begin{bmatrix} b & Ab \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

is singular.

Controllability and Reachability

Controllability Need Not Imply Reachability

So, the arbitrary state, say $[1\ 1]^\top$, can not be reached from the zero state no matter what the input is. On the other hand, any nonzero initial state can be returned to zero. For $x_1(0) = \alpha$ and $x_2(0) = \beta$, let $u(0) = -(\alpha + \beta)$, then

$$x(1) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} (-\alpha - \beta) = 0.$$

Advantages of Linear Systems :

Almost all systems present in nature are nonlinear. It is difficult to analyze a nonlinear system, so we approximate it as a linear system for our convenience and to utilize the following advantages it offers:

- Can scale and add solutions to get new solutions.
- Response to any input can be completely determined by response to a basic set of inputs, e.g., $\delta(t)$, $sin(\omega t + \phi)$, e^{-st}.
- Solution set forms a vector space. It is enough to find a basis of the solution. (Linear algebra toolkit can be used).
- The existence and uniqueness of the solution is always guaranteed.

Limitations of Linearization:

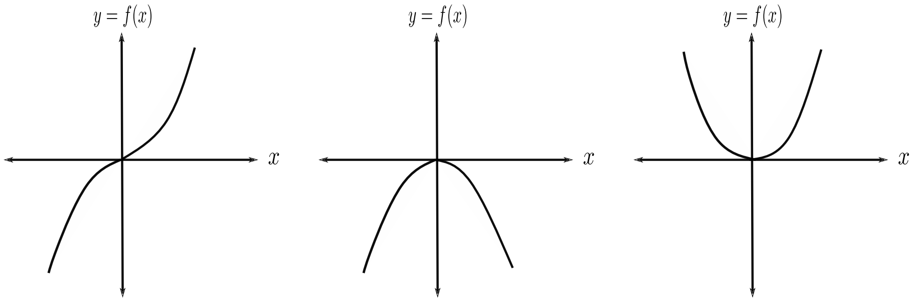

Figure: Non Linear Systems

- Linearize all three examples about the origin.
- Linearized system is given by straight line ($y = 0$).
- Systems are quite different, but linearization gives the same behaviour.

Notations

1. Set of Real no: $\mathbb{R}$
2. Let $n > 0$ is integer, set of real numbers $x_1, x_2, ...x_n$ where $x_i \in \mathbb{R}$
3. Vector or a point in $\mathbb{R}^n$ ($\mathbb{R} \times \mathbb{R} \times \mathbb{R}.... \times \mathbb{R} = \mathbb{R}^n$) space is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$$

Euclidean Space

1. Add vectors
2. Scalar multiplication
3. Zero vector in $\mathbb{R}^n$ is $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n\times 1}$. Also called a trivial vector or the origin of the $\mathbb{R}^n$ space.

Nonlinear System

Equilibrium point

Consider a nonlinear system given by

$$\dot{x} = f(t, x(t), u(t)) \qquad f : \mathbb{R}^+ \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n \tag{1}$$

A vector $x_e \in \mathbb{R}^n$ is said to be an equilibrium point of (1) if

$$f(t, x_e) = 0 \qquad \forall\ t \geq t_0, \text{ and } \quad u(t) = 0$$

Equilibrium point of LTI system

$$\dot{x} = Ax$$

- $x = 0$ is unique if A is non-singular.
- If A is singular, there will be x in null space of A. Thus, all solutions of $Ax = 0$ will be equilibrium points.

Simple Pendulum

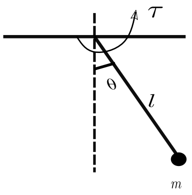

Figure: Simple Pendulum

The dynamic equation governing the equation of motion of the pendulum is given by

$$ml^2\ddot{\theta} + mgl\sin\theta = \tau$$

Consider $x_1 = \theta$ and $x_2 = \dot{\theta}$, then

Simple Pendulum

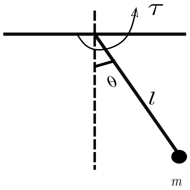

Figure: Simple Pendulum

The dynamic equation governing the equation of motion of the pendulum is given by

$$ml^2\ddot{\theta} + mgl\sin\theta = \tau$$

Consider $x_1 = \theta$ and $x_2 = \dot{\theta}$, then

$$\dot{x}_1 = x_2,$$
$$\dot{x}_2 = -\frac{g}{l}\sin x_1 + \frac{\tau}{ml^2}$$

One can write

$$\dot{x} = f(x) + g(x)u$$

Simple Pendulum

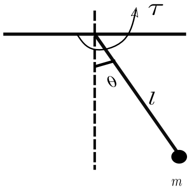

Figure: Simple Pendulum

The dynamic equation governing the equation of motion of the pendulum is given by

$$ml^2\ddot{\theta} + mgl\sin\theta = \tau$$

Consider $x_1 = \theta$ and $x_2 = \dot{\theta}$, then

$$\dot{x}_1 = x_2,$$
$$\dot{x}_2 = -\frac{g}{l}\sin x_1 + \frac{\tau}{ml^2}$$

One can write

$$\dot{x} = f(x) + g(x)u$$

where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $f(x) = \begin{bmatrix} x_2 \\ \frac{-g}{l}\sin x_1 \end{bmatrix}$, $g(x) = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix}$ and $u = \tau$.
This is affine in a control nonlinear system where $f(x)$ is a Drift vector field with $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g(x)$ is a control vector field with $g : \mathbb{R}^2 \to \mathbb{R}^2$.

Equilibrium point

$E = \{x \in \mathbb{R}^n : f(x) = 0\}$

For simple pendulum, $f(x) = 0 \implies \begin{bmatrix} x_2 \\ \frac{-g}{l} \sin x_1 \end{bmatrix} = 0$

This implies $x_2 = 0$, and $\frac{-g}{l} \sin x_1 = 0$

$x_1 = 0, \pi, 2\pi, \ldots, n\pi$

So there are two equilibrium points, $(0, 0)$, $(\pi, 0)$.

THANK YOU

Lecture 4

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Sliding mode control and its application

Linearization of non-linear system:

1. First, find the equilibrium points
2. Then, linearize the system about each equilibrium point or desired equilibrium point.

For a second order system: $\dot{x} = f(x)$, $x \in \mathbb{R}^2, f(x) \in \mathbb{R}^2$

$$\dot{x}_1 = f_1(x_1, x_2)$$
$$\dot{x}_2 = f_2(x_1, x_2)$$

Linearization about $x_e = 0$ is obtained as

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{x=x_e}$$

Let $z = x - x_e$, $\dot{z} = \dot{x} - \dot{x}_e$.
This implies the linearized system is represented as $\dot{z} = Az$

For the system $\dot{x} = f(x, u)$ where $x \in \mathbb{R}^2$ and $u \in \mathbb{R}^2$, control matrix is obtained as

$$B = \begin{bmatrix} \dfrac{\partial f_1}{\partial u_1} & \dfrac{\partial f_1}{\partial u_2} \\ \dfrac{\partial f_2}{\partial u_1} & \dfrac{\partial f_2}{\partial u_2} \end{bmatrix}_{x=x_e, u=u_e}$$

The linearized system is $\dot{z} = Az + B(u - u_e)$.

- $\dot{x} = f(x, u)$, $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$
- $\dot{x} = 0$ gives n equations, so we need to have m approximations to solve n equations for $n + m$ variables.
- These assumptions can be on the control variable or on the state variable.

Vander Pal's Equation:

$$\ddot{y}(t) - \mu(1 - y^2(t))\dot{y}(t) + y(t) = 0, \quad \text{where} \quad \mu > 0$$

Inner product

Inner product of two vectors in $\mathbb{R}^n$

$$< x, y >= x^T y = \sum_{i=1}^{n} x_i y_i$$

Length of a vector $x \in \mathbb{R}^n$

It is a norm of a vector represented as $\|x\|$ which is a non-negative real number.

$$\|.\| : \mathbb{R}^n \rightarrow \mathbb{R}^+$$

Euclidian norm:

$$\|x\|_2 \triangleq \sqrt{x_1^2 + x_2^2 + ... + x_n^2}$$

Properties of norm:

1. Positivity: $\|x\| = 0$, if and only if $x = 0$
2. Scalar multiplication: $\|\alpha x\| = |\alpha| \|x\|$, where $\alpha \in \mathbb{R}$
3. Triangle inequality: $\forall x, y \in \mathbb{R}^n$, $\|x + y\| \leq \|x\| + \|y\|$

p^{th} norm of a vector:

$1 \leq p \leq \infty, x \in \mathbb{R}^n$

$$\|x\|_p = (|x_1|^p + |x_2|^p + ... + |x_n|^p)^{1/p}$$

Infinity norm:

$$\|x\|_\infty = \max_i \{|x_i|\}, \quad i \in 1, 2, ...n$$

Distance

The notion of distance between two vectors x and y can be represented as the norm

$$\begin{aligned} d(x, y) &= \|x - y\|_2 \\ &= \sqrt{(x - y)^T (x - y)} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2} \end{aligned}$$

1-norm:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

Cauchy Schwartz inequality

Let $x, y \in \mathbb{R}^n$ then

$$|< x, y >| \leq \|x\|\|y\|$$

The inner product is a scalar quantity, so we can take its absolute value, and this absolute value satisfies the inequality above. Also, equality holds when x and y are linearly dependent.

Some relationships between different norms

- $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$
- $\|x\|_1 \leq \sqrt{n}\|x\|_2 \leq n\|x\|_\infty$

Topology definitions

Open Ball

An open ball in $\mathbb{R}^n$ of radius $r > 0$ centered around a point x, denoted by

$$B(x, r) \coloneqq \{y \in S \mid d(x, y) < r\}$$

where $S \subset \mathbb{R}^n$.

Open Set

A subset $A \subset \mathbb{R}^n$ is said to be open set if for all $x \in A$ there exists a $r_x > 0$ such that $B(x, r_x) \subset A$.

Interior point of A

Let $A \subset \mathbb{R}^n$, $x \in A$. Then x qualifies to be an interior point of A if there exists a ball $B(x, r)$ such that all points in $B(x, r)$ belongs to A.

Open Set

A set $A \subset \mathbb{R}^n$ is said to be open if all its points are interior points, $A = int(A)$.

Complement of a set

$A \subset \mathbb{R}^n$, the complement of A is $A^C = \mathbb{R}^n \setminus A$, where $\setminus$ is set minus symbol.

Adherent points/Closure points

Given a set A, $x \in \mathbb{R}^n$ is called an adherent point of A if every ball around 'x' contains at least one point from A. Formally, $\forall r > 0$, $B(x, r) \cap A \neq \emptyset$.

Bounded Set

A set 'A' is called bounded if it is contained in some ball, i.e., $A \subset B(x, r)$ for some $x \in \mathbb{R}^n$, $r > 0$.

Closed Set

The set 'A' is said to be closed if and only if it contains all its adherent points, i.e., $A = \overline{A}$, where $\overline{A}$ is set of Adherent points.

A closed set is also defined as a complement of the open set.

Limit points

A point $x \in \mathbb{R}^n$ is called a limit point of a set A if every neighborhood of 'x' contains a point of 'A' distinct from 'x'.

Closed Set

A set is said to be closed if it contains all its limit points.

Relation between adherent points and limit points

$$\overline{A} = A \cup A^{'}$$

$\overline{A}$ - closure of A, $A^{'}$- set of all limit points.

Observations

1. Every limit point is an adherent point, but the converse is not true.
2. An adherent point, which is not a limit point, is an isolated point.

Compact Set

A set 'A' is called compact if it is closed and bounded.

Boundary of a set

$$\partial A = \overline{A} \cap \overline{(\mathbb{R}^n \setminus A)}$$

- On a real line, closed intervals/sets are also compact.
- The only sets which are both open and closed are the set $\mathbb{R}^n$ and $\emptyset$.

Connnected set

A set is called disconnected if there exist two sets Y, Z such that $Y \cup Z = X$ and $\overline{Y} \cap Z = Y \cap \overline{Z} = \emptyset$.
A set is called connected if one cannot find a separation as given above, i.e., it is not disconnected.

Thank You

Lecture 5 & 6

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Sliding mode control and its application

Supremum and Infimum of set

Supremum of set:

Let S be a bounded subset of $\mathbb{R}$. A number 'b' is called the least upper bound of 'S', denoted by $b = Sup(S)$ if 'b' satisfies the following conditions:

1. 'b' is an upper bound for the set.
2. No number less than 'b' is an upper bound for S.

Infimum of a set:

Let S be a bounded set in $\mathbb{R}$. A number 'a' is said to be the greatest lower bound for the set 'S', denoted by $inf(S)$ if 'a' satisfies the following conditions:

1. 'a' is the lower bound of 'S'.
2. No number greater than a is a lower bound for S.

Continuous function

Definition of continuity

Let $f : \mathbb{R} \to \mathbb{R}$. f is said to be continuous at $x \in \mathbb{R}$, if given $\epsilon > 0$, there exists $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

If $f(.)$ is continuous at every point in $\mathbb{R}$, then 'f' is said to be continuous over $\mathbb{R}$.

For $f : \mathbb{R}^n \to \mathbb{R}^m$

$$\|x - y\| < \delta \implies \|f(x) - f(y)\| \leq \epsilon$$

Lipschitz function

Lipschitz function

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called locally Lipschitz at $x_o \in \mathbb{R}$ if there exist $(\exists)$

- a neighbourhood $B(x_o, \epsilon)$ with $\epsilon > 0$ and
- an $L > 0$

such that

$$\|f(x_1) - f(x_2)\| \leq L\|x_1 - x_2\| \qquad \text{for all}(\forall)\ \ x_1, x_2 \in B(x_0, \epsilon)$$

where L is called a Lipschitz constant. Once an L is found, anything larger also works.

Note

- Any function with a bounded first derivative must be a Lipschitz function.
- x^2, e^x and e^{-x} are few examples of functions which are locally Lipschitz on $\mathbb{R}$ but not globally Lipschitz.
- $\sin x$, $\cos x$, ax where a is constant are globally Lipschitz function.

Existence and uniqueness of solution

Existence and uniqueness for $\dot{x} = A(t)x + g(t)$:

Suppose that $A(t)$ and $g(t)$ are continuous on some open interval $J \subset \mathbb{R}$. Then, for any $t_0 \in J$ and any $x_0 \in \mathbb{R}^n$, the above equation has a unique solution satisfying $x(t_0) = x_0$.

Piece wise continuous function

There is a finite number of breaks in the function, and it does not blow to infinity.

Local existence and uniqueness for $\dot{x} = f(t, x)$:

Let $f(t, x)$ be piecewise continuous in t and satisfy the Lipschitz condition

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

$\forall\ x, y \in B = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\}\ \forall t \in [t_0, t_1]$. Then, there exists some $\delta > 0$ such that the state equation $\dot{x} = f(t, x)$ with $x(t_0) = x_0$ has a unique solution over $[t_0, t_0 + \delta]$.

For global existence and uniqueness,
$\|f(t, x) - f(t, y)\| \leq L\|x - y\| \qquad \forall\ x, y \in \mathbb{R}^n,\ \forall t \in [t_0, t_1]$.

Stability of origin

Any equilibrium point can be shifted to the origin via a change of variables.

The equilibrium point $x = 0$ of $\dot{x} = f(x)$ is

- stable if, for each $\epsilon > 0$, there is $\delta = \delta(\epsilon) > 0$ such that

$$\|x(0)\| < \delta \implies \|x(t)\| < \epsilon, \ \forall t \geq 0$$

- unstable if it is not stable.
- asymptotically stable if it is stable and $\delta > 0$ can be chosen such that

$$\|x(0)\| < \delta \implies \lim_{t \to \infty} x(t) = 0$$

Theorem 4.1 form Khalil

Let $x = 0$ be an equilibrium point for $\dot{x} = f(x)$ and $D \subset \mathbb{R}^n$ be a domain containing $x = 0$. Let $V : D \to \mathbb{R}$ be a continuously differentiable function such that

$$V(0) = 0 \quad \text{and} \quad V(x) > 0 \text{ in } D - \{0\} \tag{1}$$

$$\dot{V}(x) \leq 0 \text{ in } D. \tag{2}$$

Then, $x = 0$ is stable.
Moreover if

$$V(\dot{x}) < 0 \ in \ D - \{0\} \tag{3}$$

then $x = 0$ is asymptotically stable.

- A function $V(x)$ satisfying condition (1), that is, $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$ is said to be positive definite.
- If satisfies weaker condition $V(x) \geq 0$ for $x \neq 0$, it is said to be positive semidefinite.
- $V(x)$ is negative definite or negative semidefinite if $-V(x)$ is positive definite or positive semidefinite , respectively.

Lyapunov's Stability theorem

Lyapunov candidate function

A real symmetric matrix P is called positive definitive matrix if $x^T Px > 0 \ \forall x \neq 0$, and $x^T Px = 0$ for $x = 0$, where $x \in \mathbb{R}^n$.
The quadratic function $V = x^T Px$ is called the quadratic Lyapunov function.

Theorem for global stability

Let $x = 0$ be an equilibrium point. Let $V : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function such that,

$$V(0) = 0 \quad \text{and} \quad V(x) > 0 \forall x \neq 0 \tag{4}$$

$$\|x\| \to \infty \implies V(x) \to \infty \tag{5}$$

$$\dot{V}(x) < 0 \quad \forall x \neq 0 \tag{6}$$

then, $x = 0$ is globally asymptotically stable.

The function satisfying condition (5) is said to be radially unbounded.

Hurwitz matrix

A matrix A is Hurwitz; that is, $\mathrm{Re}\lambda_i < 0$ for all eigenvalues of A, if and only if for any given positive definite symmetric matrix Q there exists a positive definite symmetric matrix P that satisfies the Lyapunov equation

$$A^T P + PA = -Q. \tag{7}$$

Moreover, if A is Hurwitz, then P is the unique solution of (7).

The Invariance Principle

Invariant set

A set M is said to be invariant with respect to $\dot{x} = f(t, x)$ if $x(0) \in M \implies x(t) \in M$, $\forall t \in \mathbb{R}$.

This means if a solution belongs to M at some time instant, then it belongs to M for all future and past time.

Positively Invariant set

A set M is said to be positively invariant with respect to $\dot{x} = f(t, x)$ if $x(0) \in M \implies x(t) \in M, \forall t \geq 0$

LaSalle's theorem

Let $\Omega \subset D$ be a compact set that is posi tively invariant with respect to $\dot{x} = f(t, x)$. Let $V : D \to \mathbb{R}$ be a continuously differentiable function such that $\dot{V}(x) \leq 0$ in Ω. Let E be the set of all points in Ω where $\dot{V}(x) = 0$. Let M be the largest invariant set in E. Then, every solution starting in Ω approaches M as $t \to \infty$.

Corollary

Let $x = 0$ be an equilibrium point for $\dot{x} = f(t, x)$. Let $V : D \to \mathbb{R}$ be a continuously differentiable positive definite function on a domain D containing the origin $x = 0$, such that $\dot{V}(x) \leq 0$. Let $S = \{x \in D | \dot{V}(x) = 0\}$ and suppose that no solution can stay identically in S, other than the trivial solution $x(t) = 0$. Then, the origin is asymptotically stable.

Rayleigh inequality

$\lambda_{min}(P)\|x\|^2 \leq x^T Px \leq \lambda_{max}(P)\|x\|^2$

Comparison Principle

Comparison lemma

Consider the scalar differential equation

$$\dot{u} = f(t, u), \quad u(t_0) = u_0$$

where $f(t, u)$ is continuous in t and locally Liptchitz in u, for all $t \geq 0$ and all $u \in J \subset \mathbb{R}$.
Let $[t_0, T)$ (T could be infinity) be the maximal interval of existence of the solution $u(t)$, and suppose $u(t) \in J$ for all $t \in [t_0, T)$.
Let $v(t)$ be a continuous function whose upper right-hand derivative satisfies the differential inequality

$$D^+ v(t) \leq f(t, v(t)), \quad v(t_0) \leq u_0$$

with $v(t) \in J \; \forall t \in [t_0, T)$. Then, $v(t) \leq u(t)$ for all $t \in [t_0, T)$.

Upper right-hand derivative

$$D^+ v(t) = \lim_{h \to 0^+} sup \frac{v(t+h) - v(t)}{h}.$$

THANK YOU

Lecture 7 & 8

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E1 248
Sliding mode control and its application

Introduction

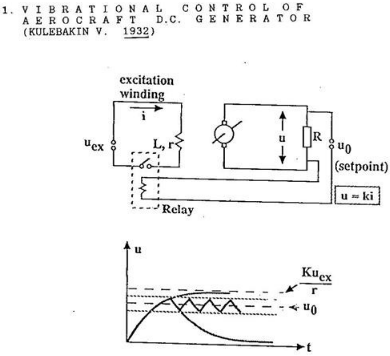


Figure: Primitive Examples - Electrical

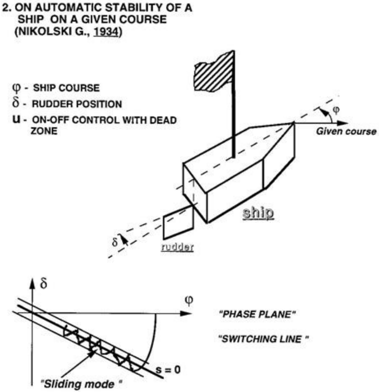


Figure: Primitive Examples-Mechanical

Sliding mode control: Revisited

Variable Structure Control: Prof. Emelyanov

- The first steps of sliding mode control theory originated in early 1950 and were initiated by S. V. Emelyanov.
- Started as Variable Structure Control (VSC):-Varying system structure for stabilization.

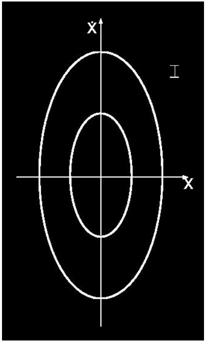


Figure: Mode I
$\ddot{x} = -k_2 x$

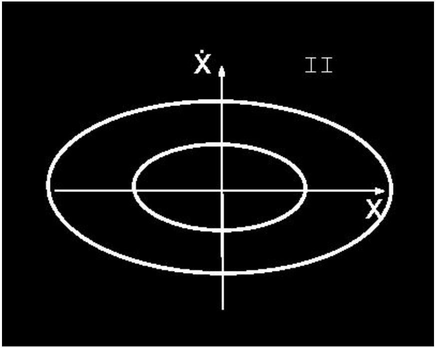
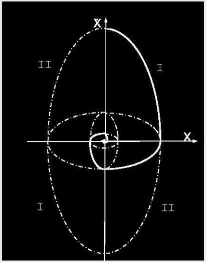


Figure: Mode II $\ddot{x} = -k_1 x$, $0 < k_1 < 1 < k_2$

Sliding mode control: Revisited

Piecing together



Properties of VSC

- Both constituent systems were oscillatory and were not asymptotically stable.
- Combined system is asymptotically stable.
- Property not present in any of the constituent systems is obtained by VSC.

Sliding mode control: Revisited

Unstable Constituent Systems

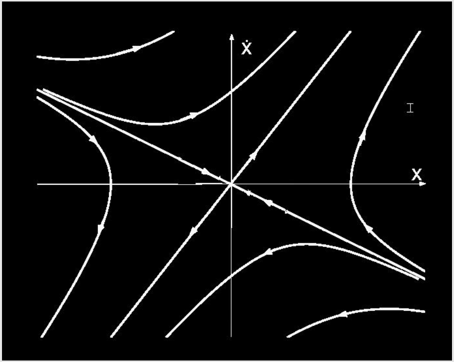


Figure: $\ddot{x} - \xi\dot{x} - \alpha x = 0$

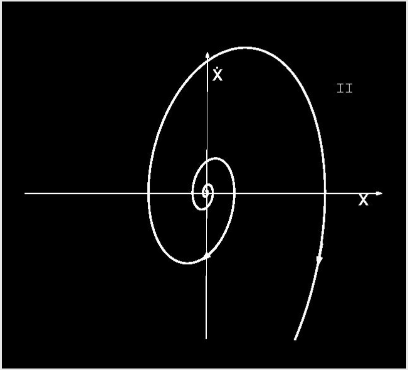


Figure: $\ddot{x} - \xi\dot{x} + \alpha x = 0$

Sliding mode control: Revisited

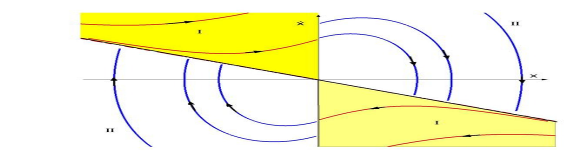
Analysis

- Both systems are unstable
- Only stable mode is one mode of the system

$$\ddot{x} - \xi\dot{x} - \alpha x = 0, \lambda = \frac{\xi}{2} - \sqrt{\frac{\xi^2}{4} + \alpha}$$

- If the following VSC is employed

$$Mode = \begin{cases} I & \text{if } xs \leq 0 \\ II & \text{if } xs > 0. \end{cases}, s = cx + \dot{x}, c = -\lambda$$



- Again, property not present in constituent systems is found in the combined system, and a stable structure can be obtained by varying between two unstable structures.

Sliding mode control design

Consider the linear system

$$\dot{x} = Ax + B(u + d(t,x))$$

$x \in \mathbb{R}^n$, $u \in \mathbb{R}$.

- The sliding surface is designed as $\sigma(x) = Cx = 0$.
- The sliding mode control is designed to reach the sliding surface in finite time.

Assumptions

- The pair (A, B), is controllable.
- The scalar $CB \neq 0$.
- The disturbance/uncertainties are matched and bounded, i.e., $|d(t,x)| \leq D_M$.

Sliding mode control design

Controller Design

Since the function $\sigma(x)$ is to be made zero, it helps to look at its derivative.

$$\begin{aligned} \dot{\sigma} &= C\dot{x} \\ &= CAx + CB(u + d(t, x)) \end{aligned} \tag{1}$$

Now, let us design the control as,

$$u = (CB)^{-1}\left(-CAx + \phi\left(\sigma\right)\right).$$

Substituting this control in (1) gives,

$$\dot{\sigma} = \phi(\sigma) + CBd(t, x) \tag{2}$$

- The function $\phi(\sigma)$ must be selected such that, the equation (2) provides finite time convergence of σ to zero, in presence of $d(t, x)$.
- Once the sliding surface is reached, i.e., $\sigma(x) \equiv 0$, the system trajectory is governed by zero dynamics, also called reduced-order dynamics.

Sliding mode control design

Examples of $\phi(\sigma)$

1. $\phi(\sigma) = -K \; sgn(\sigma)$
2. $\phi(\sigma) = -\lambda\sigma - K \; sgn(\sigma)$
3. $\phi(\sigma) = -K|\sigma|^{\alpha} \; sgn(\sigma)$

However, the third choice can not reject the disturbances if $d(t, x) \neq 0$ at $x = 0$.

Relative Degree Problem

$$\dot{\sigma} = CAx + CB(u + d(t, x))$$

- If $CB = 0$, then control does not appear in the expression for $\dot{\sigma}$.
- Usually, C is designed surface matrix so $CB = 0$ can be avoided.

However, σ may be some output of the system, and C is determined by system dynamics. The objective is to make the output zero robustly, then with $CB = 0$, the sliding mode control design is not possible.

Chattering Problem

The control is discontinuous, resulting in high-frequency switching by the actuator.

Invariance in Sliding-modes

B. Drazenovic "The invariance conditions in variable structure systems" Automatica, v.5, No.3, Pergamon Press, 1969.

Figure: Prof. B. Drazenovic

Mathematical Treatment of Matched Uncertainty

Mathematical meaning of equivalent control is that, when the trajectories are on the sliding surface then σ and $\dot{\sigma}$, both must be zero. But this assumption is valid only when the control input appears linearly for a given state. Now consider the following system

$$\dot{x} = f(x) + g(x)u + d \tag{3}$$

Invariance in Sliding modes

Consequently, one can write

$$\dot{\sigma} = \frac{\partial \sigma}{\partial x}\dot{x} = \frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} g(x) u + \frac{\partial \sigma}{\partial x} d \tag{4}$$

For calculating equivalent value of control substitute $\dot{\sigma} = 0$ in Eqn.(4), the only condition is that $\frac{\partial \sigma}{\partial x} g(x) \neq 0$, we can write

$$u_{\text{equivalent}} = -\left(\frac{\partial \sigma}{\partial x} g(x)\right)^{-1} \left(\frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} d\right) \tag{5}$$

The closed loop dynamics during sliding is obtained by substituting (5) into (3), so

$$\begin{aligned} \dot{x} &= f(x) - g(x) \left(\frac{\partial \sigma}{\partial x} g(x)\right)^{-1} \left(\frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} d\right) + d \\ &= \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x)\right)^{-1} \frac{\partial \sigma}{\partial x}\right) f(x) + \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x)\right)^{-1} \frac{\partial \sigma}{\partial x}\right) d \end{aligned} \tag{6}$$

Invariance in Sliding-modes

- Now suppose that disturbance d is entering through control channel.
- Therefore one can write $d = g(x)\delta(t)$, where $\delta(t)$ is unknown signal, but with a known bound.
- This kind of disturbance is known as matched disturbance.
- After substituting the value of d, into (6), one can write

$$\begin{aligned} \dot{x} &= \left(I - g(x)\left(\frac{\partial\sigma}{\partial x}g(x)\right)^{-1}\frac{\partial\sigma}{\partial x}\right)f(x) + \left(I - g(x)\left(\frac{\partial\sigma}{\partial x}g(x)\right)^{-1}\frac{\partial\sigma}{\partial x}\right)g(x)\delta(t) \\ &= \left(I - g(x)\left(\frac{\partial\sigma}{\partial x}g(x)\right)^{-1}\frac{\partial\sigma}{\partial x}\right)f(x) + g(x)\delta(t) - g(x)\delta(t) \\ &= \left(I - g(x)\left(\frac{\partial\sigma}{\partial x}g(x)\right)^{-1}\frac{\partial\sigma}{\partial x}\right)f(x) \end{aligned} \tag{7}$$

- Hence the closed loop dynamics, which is given in (7), is completely independent of any matched disturbances when the system is in the sliding mode.
- This property is known as the **invariance property** with respect to disturbance.

Sliding mode control design using regular form transformation

Consider the linear time-invariant system of the following form

$$\dot{x}(t) = (A + \Delta A(t))x(t) + Bu + Dd(t) \tag{8}$$

where $x \in \mathbb{R}^{n\times 1}$, $u \in \mathbb{R}^{m\times 1}$, $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$ are the state vector, control input, system matrix and input matrix respectively.
$\Delta A(t)$ and $d(t)$ represent parameter variations and disturbance vector, where D is the disturbance input matrix.

Assumptions

- The system is assumed to be controllable, i.e., the controllability matrix $[B \;\; AB \;\; A^2B \cdots A^{n-1}B]$ has full rank.
- $\Delta A \in \text{range}(B)$ and $D \in \text{range}(B)$, or there exist constant or time-varying matrices Ψ_1 and Ψ_2, such that

$$\Delta A = B\Psi_1, \quad D = B\Psi_2 \tag{9}$$

Using Eqn.(8) and assumption (9), one can write

$$\dot{x}(t) = Ax(t) + B(u + \underbrace{\Psi_1 x(t) + \Psi_2 d(t)}) = Ax(t) + B(u + \Psi_0), \;\; \Psi_0 = \Psi_1 x(t) + \Psi_2 d(t)$$

Sliding mode control design using regular form transformation

Regular Form Transformation:

Input matrix B in Eqn. (8) may be partitioned (after reordering the state vector components) as

$$B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

where $B_1 \in \mathbb{R}^{(n-m)\times m}$, $B_2 \in \mathbb{R}^{m\times m}$ with $\det B_2 \neq 0$.

The non-singular coordinate transformation T, to convert system (8) into regular form is given by

$$T = \begin{bmatrix} I_{n-m} & -B_1 B_2^{-1} \\ 0 & B_2^{-1} \end{bmatrix}$$

change of coordinate is

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = Tx$$

Sliding mode control design using regular form transformation

Transformed system (8), after applying above change of coordinate

$$\begin{aligned} \dot{z}_1(t) &= A_{11} z_1(t) + A_{12} z_2(t) \\ \dot{z}_2(t) &= A_{21} z_1(t) + A_{22} z_2(t) + u + \Psi_0 \end{aligned} \tag{10}$$

where $z_1 \in \mathbb{R}^{n-m}$, $z_2 \in \mathbb{R}^m$ and

$$TAT^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad TB = \begin{bmatrix} 0 \\ I \end{bmatrix}.$$

Design of Sliding Surface

- Now, our next aim is to design a specified switching manifold such that after reaching the manifold, the system is free from its dynamics.
- The designed manifold can be linear or nonlinear, depending on our specified goal. The simplest manifold is a linear one.
- So, here, we focus on linear manifolds only, unless otherwise specified.d.

Sliding mode control design using regular form transformation

The linear switching variable (sliding variable) is

$$S = C_1 z_1 + z_2 \tag{11}$$

where $S \in \mathbb{R}^{m\times 1}$ and $C_1 \in \mathbb{R}^{m\times(n-m)} > 0$.

Taking the first time derivative of the sliding surface (11) and substituting the value from the Eqn.(10), one can get

$$\begin{aligned}\dot{S} &= C_1\dot{z}_1 + \dot{z}_2\\ &= (C_1A_{11} + A_{21})z_1(t) + (C_1A_{12} + A_{22})z_2(t) + u + \Psi_0\end{aligned} \tag{12}$$

Expressing the system (10) into z_1 and S coordinate, using the differential Eqns.(11) and (12)

$$\begin{aligned}\dot{z}_1(t) &= A_{11}z_1(t) + A_{12}(S - C_1z_1(t))\\ \dot{S} &= (C_1A_{11} + A_{21})z_1(t) + (C_1A_{12} + A_{22})(S - C_1z_1(t)) + u + \Psi_0\\ &= (C_1A_{11} + A_{21} - C_1A_{12}C_1 - A_{22}C_1)z_1(t) + (C_1A_{12} + A_{22})S + u + \Psi_0\end{aligned} \tag{13}$$

Sliding mode control design using regular form transformation

Existence Condition of Sliding Mode:- The main objective here is to design a control u such that the sliding motion occurs in finite time. For this, the control is selected according to the following theorem

Theorem

The control input u which is defined as

$$u = \nu - (C_1A_{11} + A_{21} - C_1A_{12}C_1 - A_{22}C_1)z_1(t) - (C_1A_{12} + A_{22})S \tag{14}$$

where,

$$\nu = -K_1S - K_2\text{sign}(S))$$

with $K_1 > 0$, $K_2 > \Psi_M = \max \|\Psi_0\|$ leads to the establishment S equal to zero in finite time.

Proof

Substituting u from Eqn.(14) to (13), one can write

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}(S - C_1z_1(t)) \\ \dot{S} &= -K_1S - K_2\text{sign}(S)) + \Psi_0 \end{aligned} \tag{15}$$

Sliding mode control design using regular form transformation

Choosing the Lyapunov function as $V = \frac{1}{2}S^T S$, and calculating the time derivative of the Lyapunov function along (15), one can get

$$\begin{aligned}\dot{V} = S^T\dot{S} &= S^T(-K_1 S - K_2\text{sign}(S)) + \Psi_0) \\ &= -K_1 V - K_2\|S\| + S^T\Psi_0\end{aligned}$$

Using Cauchy-Schwartz inequality $S^T\Psi_0 \leq \|S\|\|\Psi_0\|$ and $(2V)^{\frac{1}{2}} = (S^T S)^{\frac{1}{2}} = \|S\|$, so

$$\begin{aligned}\dot{V} &\leq -K_1 V - (K_2 - \|\Psi_0\|)\|S\| \\ &\leq -K_1 V - (K_2 - \Psi_M)\|S\| \\ &= -K_1 V - 2^{\frac{1}{2}}(K_2 - \Psi_M)V^{\frac{1}{2}} \\ &= -K_1 V - \eta V^{\frac{1}{2}} < 0\end{aligned}$$

Therefore, one can ensure the asymptotic stability of the trajectory to the sliding manifold where $\eta = 2^{\frac{1}{2}}(K_2 - \Psi_M)$.

Sliding mode control design using regular form transformation

Now, we have to prove that convergence takes place in finite time.

Time calculation:-

$$\dot{V} = \frac{dV}{dt} \leq -k_1 V - \eta V^{\frac{1}{2}}$$

Use comparison lemma

$$\dot{u} = -k_1 u - \eta u^{\frac{1}{2}}$$

Then

$$T \leq \frac{2}{k_1}\left(ln\left(k_1 V(0)^{\frac{1}{2}} + \eta\right) - ln(\eta)\right)$$

Therefore, the above claim is justified since the time T is always finite.

Equivalent Control

It is clear from (14), during the sliding mode $S = 0$ and the average value of $\nu = 0$, therefore the remaining control is $u_{\text{equivalent}} = -(C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1) z_1(t)$, which is theoretically interpreted as **equivalent control**, which is the control required to maintain a sliding motion on the manifold S.

Sliding mode control design using regular form transformation

Analysis of Reduced Order Dynamics

- Therefore, after time T, dynamics of S have been collapsed, and the reduced order dynamics of the system (15) can be obtained by substituting $S = 0$.
- The reduced order dynamics is given as

$$\dot{z}_1(t) = (A_{11} - A_{12}C_1)z_1(t)$$

- The stability and performance of the above reduced order system depend on (A_{11}, A_{12}).
- Therefore, the design of C_1 depends on the controllability of the pair (A_{11}, A_{12}).

Lemma

The matrix pair (A_{11}, A_{12}) is controllable, if and only if the pair (A, B) is controllable.

Sliding mode control design using regular form transformation

Proof

$$\begin{aligned}\text{rank}[sI - A \quad B] &= \text{rank}\begin{bmatrix} sI - A_{11} & -A_{12} & 0 \\ -A_{21} & sI - A_{22} & I \end{bmatrix} \\ &= \text{rank}\begin{bmatrix} sI - A_{11} & A_{12} \end{bmatrix} + m \quad \text{for all } s \in \mathbb{C}\end{aligned}$$

This implies

$$\text{rank}[sI - A \quad B] = n \Leftrightarrow \text{rank}\left[sI - A_{11} \quad A_{12}\right] = n - m$$

and from the Popov-Belevitch-Hautus (PBH) rank test, it follows that (A, B) is controllable if and only if the pair (A_{11}, A_{12}) is controllable.

Now, one can design C_1 using any robust linear state feedback method, such as quadratic minimization, direct or robust eigenvalue assignment, using LMI, etc.

Thank You!

Lecture 8 & 9

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E1 248
Sliding mode control and its application

Outline

Introduction

Consider autonomous systems of the form

$$\dot{x}(t) = Ax(t) \tag{1}$$

where $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ to formulate the **concepts of algorithm.**

If the eigenvalues of matrix A are in the left half s-plane, then one can easily prove that the system is asymptotically stable.

Now, let us consider a non-autonomous system of the following form

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x \in \mathbb{R}^n, \quad A \in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^n, \quad u \in \mathbb{R}. \tag{2}$$

- Our objective is to design a control u, so that (2) is converted to an autonomous system like (1).
- If we are able to do this, then one can claim that the system (2) has similar properties as (1). In other words, one can say that (1) is an **algorithm**.
- **Therefore, here, the meaning of algorithm is a set of autonomous differential equations/equations which have the desired properties.**

Introduction

- The simplest form of state-dependent controller for the system (2) is $u = Kx$.
- With this control the closed loop system (2) is $\dot{x}(t) = (A + BK)x(t)$.
- Therefore, now we have to select K such that the new system matrix $(A + BK)$ is Hurwitz.

Remark:

- Any non-autonomous system is converted to a specific algorithm, which has the desired properties, by substituting an appropriate form of control.
- Therefore, the controller design is always dependent on the algorithm.

- In the above analysis, we are considering only nominal systems without any external disturbances/perturbations.
- But almost all practical systems are influenced by external disturbances/ perturbations.

Analysis of perturbed systems

Let a perturbed autonomous system be represented as

$$\dot{x}(t) = Ax(t) + d \tag{3}$$

where states $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ is Hurwitz system matrix.

Assumptions

- The disturbance $d = d_1 + d_2 x \in \mathbb{R}^n$.
- d_1 is the non vanishing disturbance which is bounded as $d_{1L} \leq \|d_1\| \leq d_{1U}$.
- d_2 is the vanishing disturbance which is bounded as $d_{2L} \leq \|d_2\| \leq d_{2U}$.

- Before going to further analysis, let us define the types of disturbances that frequently affect any deterministic systems.
- Deterministic systems represent that class of system where the disturbance affecting is bounded, but the actual value of disturbance is unknown.

Types of disturbances

Vanishing Disturbance:

- The disturbance which vanishes at the equilibrium point(origin), i.e., if $d = 0$, at the equilibrium point, then the disturbance is called vanishing disturbance.
- In such a case, the equilibrium point of the perturbed system (3) is the same as that of the nominal system (1).
- This means the equilibrium point is preserved, and one can analyze the stability with respect to the origin.

Non-vanishing Disturbance:

- If $d \neq 0$, at equilibrium point(origin), then the disturbance is called non-vanishing disturbance.
- In this case equilibrium point of perturbed system (3) is not same as the nominal system (1).
- Therefore, one can no longer study the problem as a question of stability of equilibria.
- Some new formulation is required to study these classes of systems.

Stability analysis of systems

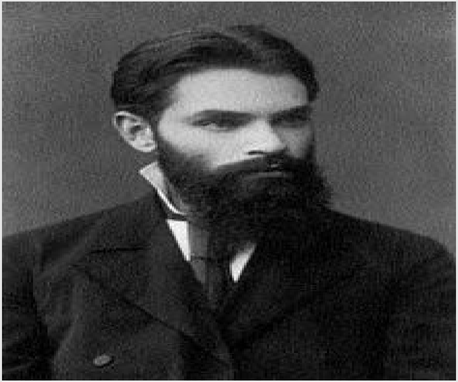

Figure: Aleksandr Lyapunov

Stability analysis of systems

- There are several ways of analyzing the stability of the dynamical systems, Lyapunov stability is one among them.
- In Lyapunov stability, an energy-like function is defined,
- This function is always positive and whose value is zero at the equilibrium point, and also, the derivative of this energy-like function is monotonically decreasing or at least non-increasing along the system trajectory.

Mathematically, one can write

$$\begin{aligned} V(x) &> 0 \\ V(x=0) &= 0 \\ \dot{V}(x) &< 0 \end{aligned}$$

where $V(x)$ is the positive definite energy-like function.

Remark:

Lyapunov function might also be explicitly time-dependent. But here, we are considering only state-dependent Lyapunov functions.

Stability analysis of systems

The most frequently used Lyapunov candidate function is a quadratic Lyapunov function. The quadratic Lyapunov function for the system (3) is given by

$$V(x) = x^T P x$$

where $P = P^T$ is a positive definite symmetric matrix.

Taking the time derivative of the Lyapunov function along the system trajectories (3), we obtain,

$$\begin{aligned} \dot{V}(x) &= \dot{x}^T P x + x^T P \dot{x} \\ &= x^T A^T P x + x^T P A x + d^T P x + x^T P d \\ &= x^T (A^T P + PA) x + d^T P x + x^T P d \end{aligned}$$

Let $Q = Q^T > 0$ be another positive definite symmetric matrix such that,

$$A^T P + PA = -Q$$

Stability analysis of systems

Cauchy-Schwartz inequality

$$|x^T P d| \leq \lambda_{max}(P)\|x\|\|d\|$$
$$|d^T P x| \leq \lambda_{max}(P)\|x\|\|d\|$$

where $\|.\|$ represents the norm

Rayleigh inequality

$$\lambda_{min}(P)\|x\|^2 \leq x^T P x \leq \lambda_{max}(P)\|x\|^2$$
$$\lambda_{min}(Q)\|x\|^2 \leq x^T Q x \leq \lambda_{max}(Q)\|x\|^2$$

where $\lambda_{min}(.)$ and $\lambda_{max}(.)$ represent the minimum and maximum eigenvalues of the matrices, respectively. Using above inequalities

$$\begin{aligned} \dot{V}(x) &= x^T(-Q)x + d^T P x + x^T P d \\ &\leq -\lambda_{min}(Q)\|x\|^2 + 2\lambda_{max}(P)\|x\|\|d\| \end{aligned} \tag{4}$$

Stability analysis of systems

Let us consider the case of vanishing disturbance, $d = d_2 x$ or $\|d\| \leq d_{2U}\|x\|$. Substituting the value of $\|d_2\|$ in Eqn.(4)

$$\begin{aligned}\dot{V}(x) &\leq -\lambda_{min}(Q)\|x\|^2 + 2\lambda_{max}(P)d_{2U}\|x\|^2 \\ &\leq -(\lambda_{min}(Q) - 2\lambda_{max}(P)d_{2U})\|x\|^2\end{aligned}$$

Hence origin is globally asymptotically stable if $d_{2U} < \frac{\lambda_{min}(Q)}{2\lambda_{max}(P)}$ which satisfies the condition $\dot{V}(x) < 0$.

Now consider the case of non vanishing disturbance $d = d_1$ or $\|d\| \leq d_{1U}$. Substituting the value of $\|d_1\|$ in Eqn.(4)

$$\begin{aligned}\dot{V}(x) &\leq -\lambda_{min}(Q)\|x\|^2 + 2\lambda_{max}(P)d_{1U}\|x\| \\ &\leq -(\lambda_{min}(Q)\|x\| - 2\lambda_{max}(P)d_{1U})\|x\|\end{aligned}$$

When states of the system are outside the set $\Omega = \left\{x \in \mathbb{R}^n : \|x(t)\| < \frac{2\lambda_{max}(P)d_{1U}}{\lambda_{min}(Q)}\right\}$, then $\dot{V} < 0$. Hence system trajectories are **ultimately bounded in the set Ω.**

Some more analysis of systems with a non-vanishing disturbance

- From the above analysis, it is clear that non-vanishing disturbances have to be dealt with in different ways.
- But one can argue that the system can be stabilized easily with the help of a control variable.
- Now, let us suppose that we have a control input also in the system $\dot{x}(t) = Ax(t) + d$.

For simplicity, we consider a system with the number of states variables equal to the number of control inputs or mathematically $B \in \mathbb{R}^{n \times n}$ and $u \in \mathbb{R}^n$

$$\dot{x}(t) = Ax(t) + Bu(t) + d \tag{5}$$

It can be seen that even with the stated condition, stabilization of the system (5) is not easy, with a simple continuous controller.

Some more analysis of systems with a non-vanishing disturbance

Suppose we are taking a state feedback-based control assuming that all the state variables are available and the system is completely controllable.

After applying the state feedback control $u = Kx(t)$, the closed loop system (5) becomes

$$\begin{aligned} \dot{x}(t) &= Ax(t) + BKx(t) + d \\ &= (A + BK)x(t) + d \end{aligned} \tag{6}$$

Eqn. (6) is again the same as (3) with a non-vanishing disturbance, and again similar kind of situation arises, and it is difficult to stabilize the system at the origin.

This problem can be solved with the **high gain feedback** theory.

In high gain feedback, the control is designed as $u = \lim_{\tau \to 0} \frac{1}{\tau} Kx(t)$, where $K < 0$ and $\tau \to 0^+$.

High gain feedback

After applying high gain feedback the closed loop system (5) can be represented as

$$\dot{x}(t) = Ax(t) + \lim_{\tau \to 0} \frac{BK}{\tau} x(t) + d$$

On simplification,

$$\tau \dot{x}(t) = \tau Ax(t) + BKx(t) + \tau d \Rightarrow BKx(t) \approx 0 \tag{7}$$

because $\tau \dot{x}(t) = \tau Ax(t) = \tau d \approx 0 \Rightarrow x \to 0$ if BK is invertible.

Observations:

- Hence, mathematically, one can see that despite non-vanishing disturbances, the system can be stabilized at the origin.
- But this control has infinite magnitude, therefore practically it is not feasible to apply this type of control because of the finite bandwidth of all practical actuators.
- Again, the problem of non-vanishing disturbance remains open from the practical point of view, but at least a good mathematical insight is developed.

High gain feedback for more generalized system

Consider the following system

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}z_2(t) \\ \dot{z}_2(t) &= A_{21}z_1(t) + A_{22}z_2(t) + u + \Psi_0 \end{aligned} \tag{8}$$

High gain feedback control is designed as
$u = -A_{21}z_1(t) - A_{22}z_2(t) + \lim_{\tau\to 0}\frac{1}{\tau}\left(K_1z_1(t) + K_2z_2(t)\right)$ for the system (8)

After applying high gain feedback, the closed loop system (8) can be represented as

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}z_2(t) \\ \dot{z}_2(t) &= \lim_{\tau\to 0}\frac{1}{\tau}\left(K_1z_1(t) + K_2z_2(t)\right) + \Psi_0 \end{aligned}$$

One can further write

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}z_2(t) \\ \lim_{\tau\to 0}\tau\dot{z}_2(t) &= \left(K_1z_1(t) + K_2z_2(t)\right) + \lim_{\tau\to 0}\tau\Psi_0 \end{aligned}$$

High gain feedback for more generalized system

$$
\begin{aligned}
\Rightarrow \dot{z}_1(t) &= A_{11} z_1(t) + A_{12} z_2(t) \\
K_1 z_1(t) &+ K_2 z_2(t) \approx 0
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \dot{z}_1(t) &\approx A_{11} z_1(t) - \frac{A_{12} K_1}{k_2} z_1(t) \\
&\approx \left(A_{11} - \frac{A_{12} K_1}{K_2} \right) z_1(t)
\end{aligned}
$$

which implies $z_1 \approx 0$ if poles of $A_{11} - \frac{A_{12} K_1}{K_2}$ lies in left half plane.

Since $K_1 z_1(t) + K_2 z_2(t) \approx 0$ and $z_1(t) \approx 0$, which further implies $z_2(t) \approx 0$.

Continuous to Discontinuous Feedback Controller

- It is also observed that after applying such a high gain feedback control, the system order is reduced, which is clearly seen from Eqn. (7).
- A Vast amount of research was carried out by many to explore the possibility of having the same phenomenon as the above with a finite magnitude of control.
- The only unexplored area was controlling systems using discontinuous feedback controllers, which will expand the scope of selecting the controllers.
- Enormous research in this direction led to the development of a discontinuous algorithm for the continuous autonomous system, which can be stabilized at the origin despite non-vanishing disturbances.

Discontinuous autonomous systems without disturbance

Let us take the case of the following simplest disturbance free one-dimensional discontinuous autonomous system

$$\dot{x}(t) = -k\text{sign}(x(t)) \tag{9}$$

where $x \in \mathbb{R}$, $k > 0$ and

$$\text{sign}(x(t)) = \begin{cases} +1 & \text{if } x(t) > 0 \\ \{-1 \quad 1\} & \text{if } x(t) = 0 \\ -1 & \text{if } x(t) < 0 \end{cases}$$

Remark:

- One can observe that Eqn.(9) is not a simple differential equation,
- It contains an infinite number of differential equations because of the infinite number of possibilities at $x(t) = 0$ (differential equation of right-hand side can take any value between $\{$-1 1$\}$).
- This type of equation is named differential equations with discontinuous right-hand side or more soundly **differential inclusion**.
- The existence of a solution of this type of differential equation with a discontinuous right-hand side is not explored in the classical sense.

Discontinuous autonomous systems without disturbance

Taking the simplest candidate Lyapunov $V = \frac{1}{2}x^2(t)$ for the system (9) and taking the time derivative of Lyapunov function along the system trajectory, we obtain

$$\begin{aligned} \dot{V} &= x(t)\dot{x}(t) \\ &= x(t)(-k\text{sign}(x(t))) \\ &= -k|x(t)| < 0 \end{aligned} \tag{10}$$

because $x(t)\text{sign}(x(t)) = |x(t)|$.

- Hence, one can conclude from the above analysis that the system (9) is asymptotically stable at the origin.
- But, we can also see that the above analysis can be extended like

$$\dot{V} = -k(2V)^{\frac{1}{2}} \tag{11}$$

because $V = \frac{1}{2}(x(t))^2 = \frac{1}{2}(|x(t)|)^2$.

Discontinuous autonomous systems without disturbance

One can solve Eqn.(11) as

$$\frac{dV}{dt} = -k(2V)^{\frac{1}{2}} \Rightarrow \int_{V(0)}^{0} \frac{dV}{(V)^{\frac{1}{2}}} = -k2^{\frac{1}{2}} \int_{0}^{T} dt$$
$$\frac{(V(0))^{\frac{1}{2}}}{\frac{1}{2}} = k2^{\frac{1}{2}} T \Rightarrow T = \frac{(2V(0))^{\frac{1}{2}}}{k} \tag{12}$$

where $V(0)$ is initial value of Lyapunov function and T is final convergence time to the origin.

Note:

- We have already proved that the system is asymptotically stable, which means that the system states are attracted towards the origin,
- Therefore, the value of the Lyapunov function also decreases along the trajectory,
- Hence, we are taking the lower limit of integration $V = V(0)$ at $t = 0$, and finally, the system converges towards the origin, so the upper limit of integration is $V = 0$ at $t = T$.

Discontinuous autonomous systems with a non-vanishing disturbance

Remark:

- From the above analysis, it is clear that the system states converge to the origin in a finite time $T = \frac{(2V(0))^{\frac{1}{2}}}{k}$.
- Therefore, the system (9) is finite time stable, which has a quicker convergence than an asymptotic stable system like (1).

- Now we come to our main problem of interest, system (9) with a non-vanishing bounded disturbance $|d| \leq d_{max} \in \mathbb{R}$ where d_{max} is the maximum value of the disturbance magnitude.
- Our task is to analyze the stability of the perturbed system, which is expressed as

$$\dot{x}(t) = -k\text{sign}(x(t)) + d \tag{13}$$

Discontinuous autonomous systems with a non-vanishing disturbance

Again choosing the same candidate Lyapunov $V = \frac{1}{2}x^2(t)$ for the system (13) and taking its time derivative along the system trajectory, we can write

$$\dot{V} = x(t)\dot{x}(t) = x(t)(-k\text{sign}(x(t)) + d) = -k|x(t)| + dx(t) \tag{14}$$

Now using norm inequality,

$$dx(t) \leq |d||x(t)| \leq d_{max}|x(t)| \tag{15}$$

Using Eqn.(14) and (15),

$$\begin{aligned} \dot{V} &\leq -k|x(t)| + d_{\max}|x(t)| \\ &\leq -(k - d_{\max})|x(t)| \\ &\leq -\eta(2V)^{\frac{1}{2}} \end{aligned} \tag{16}$$

where $\eta \geq (k - d_{\max}) > 0$. Therefore, if $k \geq d_{max}$ then the system (13) is finite time stable. By comparison lemma again one can calculate the time $T \leq \frac{(2V(0))^{\frac{1}{2}}}{\eta}$.

Open problem during early fifties

Consider a generic mechanical system given by following sets of differential equation

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u + f(t, x) \\ \sigma &= x_2 \end{aligned} \tag{17}$$

where $x_1 x_2, u, f(t, x)$ and σ are the position, velocity, control input, all other nonlinear dynamics with perturbation/unmodelled dynamics, and the measured output, respectively.

Now, if we select control as

$$u = -k\text{sign}(\sigma) \tag{18}$$

The above control is able to stabilize $x_2 = 0$ in finite time, same as (13), mathematically when $x_2 = 0 \Rightarrow x_1 = c$, where c is an arbitrary constant.

Open problem during early fifties

The open problems are

- The system is stopped, but where?
- Can we manipulate both x_1 and x_2 at the same time?
- A definition of solution at the discontinuity surface is needed.
- High frequency discontinuous (switching) control.
- Chattering.

Problem formulation

Design $u(t)$ such that

$$\lim_{t \to \infty} x_1(t) = \lim_{t \to \infty} x_2(t) = 0 \tag{19}$$

in the presence of bounded uncertainty, i.e., $|f(t,x)| < L$, where $f(t,x)$ represents the modeling imperfections and external perturbations and L represents the maximum bound of uncertainty.

Open problem during early fifties

Remarks:-

- As we already discussed that system of the form $\dot{x}(t) = u + d$, where $u = -k\text{sign}(x(t)) \in \mathbb{R}$ is able to stabilize x only when $x \in \mathbb{R}$.
- But in practical scenarios, almost all systems have more number of states than number of input.
- If the number of free variables to be controlled is equal to the number of controlled inputs, then the above-proposed controller will work well.
- But, when one can try to apply the same algorithm for the vector systems with a scalar control, then further concepts of the manifold are needed. The required manifold in this case is known as **sliding surface**.
- Therefore, the sliding manifold is one of the ways by which we can control any vector system using scalar control.

Founder of Sliding Mode

Prof. Utkin and Prof. Emel'yanov, IFAC Sensitivity Conference, Dubronovik 1964

Figure: Father of Sliding Mode Control Theory

Concept of Manifold

- Although, a k-dimensional manifold in $\mathbb{R}^n (1 \leq k < n)$ has a rigorous mathematical definition.
- But for our purpose, it is sufficient to think of a k-dimensional manifold as the solution of the equation
$$\sigma(x) = 0$$
where $\sigma : \mathbb{R}^n \to \mathbb{R}^{n-k}$ is sufficiently smooth (that is, sufficiently many times continuously differentiable).

Example:

The unit circle
$$\left\{ x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1 \right\}$$
is a 1-dimensional manifold in $\mathbb{R}^2$.

Example:

The unit sphere
$$\left\{ x \in \mathbb{R}^n \mid \sum_{1}^{n} x_i^2 = 1 \right\}$$
is an $(n-1)$-dimensional manifold in $\mathbb{R}^n$.

Concept of Manifold

Example:-Consider the following uncertain chain of integrators

$$\begin{aligned}\dot{x}_1 &= x_2\\ &\vdots\\ \dot{x}_{n-1} &= x_n\\ \dot{x}_n &= f(x) + g(x)u\end{aligned} \tag{20}$$

- Here, $\sigma(x) = \sum_{i=1}^{n} c_i x_i = 0$, (where $c_i > 0, i = 1, 2 \cdots n$ are the coefficients of the Hurwitz matrix) is $(n-1)$ dimensional manifold in $\mathbb{R}^n$.
- In the first order sliding mode $\sigma(x) = 0$ is commonly taken as a manifold.
- But manifold can be nonlinear also.

Let us now define a sliding manifold.

Definition of Sliding Manifold

The chosen line/plane/surface of the state space along which the motion of the system trajectory occurs after finite time is known as a sliding manifold.

Problem of Interest

Consider the nonlinear uncertain system of the following form

$$\dot{x} = f(x) + g(x)u \tag{21}$$

where $x \in X \in \mathbb{R}^{n\times 1}$ the state vector and $u \in \mathbb{R}$ the control input.

Assumptions:

- Functions $f(x)$ and $g(x)$ are smooth uncertain functions and are bounded for X
- $f(x)$ contains unmeasured perturbation terms and $g(x) \neq 0$ for $x \in X$
- System (21) is controllable for all $x \in X$.
- Continuous function σ (named as a sliding variable) admits a relative degree equal to 1 with respect to u.
- $|\Psi| \leq |\Psi_M|$ and $0 < \Gamma_m \leq \Gamma \leq \Gamma_M$ for $x \in X$. It is assumed that Ψ_M, Γ_m and Γ_M exist and known.

Aim -**The control objective consists in forcing the continuous function $\sigma(x,t)$, to 0 in finite time.**

Definition of Sliding Mode

Taking time derivative of sliding variable $\dot{\sigma}$ along the system (21), one can write

$$\dot{\sigma} = \frac{\partial \sigma}{\partial x}\dot{x} + \frac{\partial \sigma}{\partial t} = \underbrace{\frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial t}} + \frac{\partial \sigma}{\partial x} g(x) u$$
$$= \Psi(x,t) + \Gamma(x,t) u \tag{22}$$

where $\Psi(x,t) = \frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial t}$ and $\Gamma(x,t) = \frac{\partial \sigma}{\partial x} g(x)$.

Definition:

- Consider the nonlinear system (21) and let the system be closed by some possibly dynamical discontinuous feedback.
- Variable σ is a continuous function and the set

$$S = \{x \in X | \sigma(x,t) = 0\} \tag{23}$$

called sliding surface, is non-empty and is locally an integral set in the Filippov sense, i.e., it consists of Filippov's trajectories of the discontinuous dynamical system.
- The motion on S is called sliding mode with respect to the sliding variable σ.

Definition of Sliding Mode

Remark :

In simple words, one can define the first-order ideal sliding mode algorithm as finite-time stabilization of uncertain bounded integrator (22) using discontinuous feedback (with infinite switching frequency) in the sense of Filippov.

Real Sliding Mode

Given the sliding variable $\sigma(x, t)$, the real sliding surface associated with (21) is defined as (with $\delta > 0$)

$$S^* = \{x \in X| \quad |\sigma| < \delta\}. \tag{24}$$

Definition :

- Consider the non-empty real sliding surface S^* given by (24),
- and assume that it is locally an integral set in the Filippov sense.
- The corresponding behavior of the system (21) on (24) is called real sliding mode with respect to the sliding variable $\sigma(x, t)$.

Analysis of Sliding Motion

Phases of Sliding Mode Control

- **Reaching phase**: Where the system state is driven from any initial state to reach the switching manifolds (the anticipated sliding modes) in finite time.
- **Sliding-mode phase**: Where the system is induced into the sliding motion on the switching manifolds, i.e., the switching manifolds become an attractor.

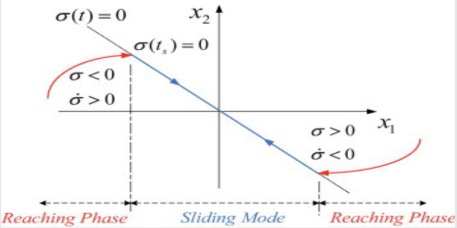

Figure: Phases of Sliding Mode Control

Physical Realization of SMC

Design Steps of Sliding Mode Control

- **Switching manifold selection**: A set of switching manifolds are selected with prescribed desirable dynamical characteristics.
- **Discontinuous control design**: A discontinuous control strategy is formed to ensure the finite time reachability of the switching manifolds. The controller may be either local or global, depending upon specific control requirements.

Finite Time Reachability to the Sliding Manifold

- Consider the autonomous differential equation described by

$$\dot{x} = f(x) \tag{25}$$

where, $x \in \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^n$ is continuous on an open neighbourhood $\mathcal{D}$ of the origin and locally lipschitz on $\mathcal{D} \setminus 0$.

- Assuming the origin is the only equilibrium point of (25) following are some results for finite time stability of the origin given in next slide.

Finite Time Stability

Definition:

Suppose there exists a continuous function $V : \mathcal{D} \to \mathbb{R}$ such that the following conditions hold.

1. V is positive definite.
2. $\dot{V}$ is continuous and negative on $\mathcal{D} \setminus 0$
3. There exists real positive numbers c and α such that $\dot{V} + cV^{\alpha} \leq 0$ on $\mathcal{D} \setminus 0$.

then the origin of (25) is finite-time stable.

Above result is a special case of more general result available in literature as follows.

Suppose there exists a continuous function $V : \mathcal{D} \to \mathbb{R}$ such that the following conditions hold.

1. V is positive definite.
2. $\dot{V} \leq \phi(V)$, and using comparison principle $\dot{v} = \phi(v)$, with $v \in \mathbb{R}$ and $V(x_0) \leq v(0)$ is a finite time stable differential equation.

then the origin of (25) is finite-time stable.

Reaching Law

Synthesizing the reaching law:

- Let $V = \frac{1}{2}\sigma^2$ be the Lyapunov function for synthesizing the reaching law, where $\sigma \in \mathbb{R}$ is the sliding surface.
- The time derivative of the Lyapunov function is given as

$$\dot{V} = \sigma\dot{\sigma} \tag{26}$$

- For the asymptotic stability, $\dot{V} < 0 \Rightarrow \sigma\dot{\sigma} < 0$.
- But the objective here is to reach the sliding surface in finite time, for this purpose, if $\dot{V}(x) + cV^{\alpha}(x)$ is negative semidefinite,
- , then one can ensure the finite time reachability to the surface.
- There are infinite number of solutions satisfying the inequality $\dot{V}(x) + cV^{\alpha}(x) \leq 0$.
- One of the possibilities is

$$\sigma\dot{\sigma} \leq -\eta|\sigma| \tag{27}$$

where $\eta > 0$ which is known as $\eta-$**reachability condition**, which ensures finite time convergence to $\sigma = 0$,

Reaching Law

Condition for Convergence

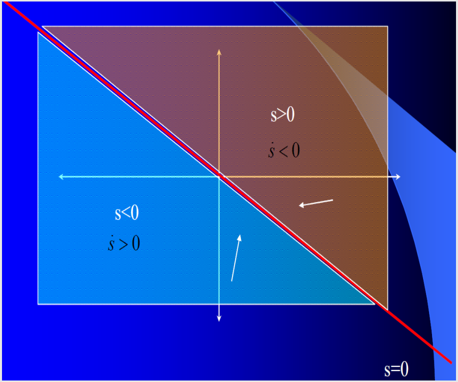

Figure: Condition for Convergence

Analysis of Reachability Condition

- By integration of (27)

$$|\sigma(t)| - |\sigma(0)| \leq -\eta t \tag{28}$$

showing that the time required to reach the surface, starting from initial condition $\sigma(0)$ is bounded by

$$T = \frac{\sigma(0)}{\eta} \tag{29}$$

- This possibility comes into picture by putting $\alpha = \frac{1}{2}$ into the condition $\dot{V}(x) + cV^{\alpha}(x) \leq 0$.
- One of the solution (27) is $\dot{\sigma} = -k\text{sign}(\sigma)$, $k > 0$, which is known as **constant reaching law**.

Problem with constant reaching law

- The reaching rate is directly dependent on the magnitude of the gain.
- But, the main problem with a large gain is that the chattering magnitude is directly proportional to the gain of the discontinuous term.

Analysis of Reaching Law

Problem with constant reaching law and simple modification

- If $k = k_1 + k_2|\sigma|$, $k_1 > 0, k_2 > 0$, the convergence speed of the trajectory towards the sliding surface,
- when the initial state is far away from the sliding surface $k_2|\sigma|$ will dominate k_1, so $k \approx k_2|\sigma|$
- and also chattering magnitude is reduced because $k_2|\sigma| = 0$, at $\sigma = 0$.

The above modification in constant rate reaching law is mathematically written as

$$\begin{aligned}\dot{\sigma} &= -k\text{sign}(\sigma) \\ &= -(k_1 + k_2|\sigma|)\text{sign}(\sigma) = -k_2\sigma - k_1\text{sign}(\sigma)\end{aligned} \tag{30}$$

Eqn. (30) is known as **proportional rate reaching law**.
This satisfies the $\eta-$ reachability condition

$$\begin{aligned}\sigma\dot{\sigma} &= \sigma(-k_2\sigma - k_1\text{sign}(\sigma)) \\ &= -k_2\sigma^2 - k_1|\sigma| \leq -k_1|\sigma|\end{aligned} \tag{31}$$

More Discussion on $\eta-$Reachability Condition

- $\eta-$reachability is only a sufficient but not a necessary condition for the finite time reachability to the sliding surface.
- One can see the following power rate reaching law (32), which has the property of finite time reachability to the sliding surface but does not satisfy the $\eta-$reachability condition.

$$\dot{\sigma} = -k|\sigma|^{\alpha}\text{sign}(\sigma), \quad 0 < \alpha < 1 \tag{32}$$

- Checking the $\eta-$reachability condition

$$\sigma\dot{\sigma} = -k|\sigma|^{1+\alpha} \nleq -k|\sigma| \tag{33}$$

when $0 < \sigma < 1$ then, $-k|\sigma|^{1+\alpha} \nleq -k|\sigma|$. But (32) is finite-time stable.

For justifying this statement mathematically take the Lyapunov function as $V = |\sigma|$. Taking the time derivative of this Lyapunov function along the system (32), one can write

$$\dot{V} = \dot{\sigma}\text{sign}(\sigma) = -k|\sigma|^{\alpha} = -kV^{\alpha} \tag{34}$$

Hence the above claim is justified.

Advantages of Sliding Mode Control

Advantages:

- Exact compensation (insensitivity) with respect to bounded matched uncertainties.
- Reduced order dynamics during sliding.
- Finite-time convergence to the sliding surface.

Above advantages can be illustrated using the following example of single-input-single-output systems with motion equations in canonical space

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= -\sum_{i=1}^{n} a_i(t)x_i + b(t)u + f(t) \end{aligned} \tag{35}$$

where $a_i(t)$ and $b_i(t)$ are unknown parameters and $f(t)$ is an unknown disturbance. Here $x = x_1$ is a controlled variable, and its time derivatives $x^{i-1} = x_i, i = 1, 2, \cdots, n$ are components of a state vector in the canonical space.).

Advantages of Sliding Mode Control

For the system (35) switching manifold is designed as $\sigma(x) = \sum_{i=1}^{n} c_i x_i = 0$, (where $c_i > 0, i = 1, 2 \cdots n$ are the coefficients of the Hurwitz matrix. Now taking the time derivative of the sliding manifold, one can write

$$\begin{aligned}
\dot{\sigma} &= c_1\dot{x}_1 + c_2\dot{x}_2 + \cdots + c_n\dot{x}_n \\
&= c_1x_2 + c_2x_3 + \cdots + c_n\left(-\sum_{i=1}^{n} a_i(t)x_i + b(t)u + f(t)\right)
\end{aligned} \tag{36}$$

For simplicity take $c_n = 1$. Transforming the system (35) in the coordinate of σ gives

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
&\vdots \\
\dot{x}_{n-1} &= \sigma - \sum_{i=1}^{n-1} c_i x_i \\
\dot{\sigma} &= c_1x_2 + c_2x_3 + \cdots - \sum_{i=1}^{n-1} a_i(t)x_i - a_n\left(\sigma - \sum_{i=1}^{n-1} c_i x_i\right) + b(t)u + f(t)
\end{aligned} \tag{37}$$

Advantages of Sliding Mode Control

After choosing the proper value of switching controller u, $\sigma = 0$ and dynamics of $\dot{\sigma}$ is collapsed in finite time.

Hence reduced-order dynamics is given as

$$\dot{x}_1 = x_2$$
$$\vdots$$
$$\dot{x}_{n-1} = -\sum_{i=1}^{n-1} c_i x_i \tag{38}$$

which is free from disturbances and system uncertainties.

Thank You!

Lecture 10

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E1 248
Sliding mode control and its application

Outline

Equivalent control method

Equivalent control method by the geometrical point of view

- Interpretation of the control signal in terms of its average or low-frequency behavior.
- The purpose of the control action is to ensure that the trajectories are driven towards and forced to remain on the sliding surface to guarantee a sliding motion.
- From a geometrical point of view, the equivalent control method does the same job. It replaces the discontinuous control on the intersection of the switching surface with a continuous one.

Mathematically, consider the system

$$\dot{x} = f(x) + B(x)u, \quad \text{where} \quad x,\ f(x) \in \mathbb{R}^n, \quad B(x) \in \mathbb{R}, \quad u \in \mathbb{R}^n \tag{1}$$

Equivalent control method

Control input is defined as

$$u(x) = \begin{cases} u^+ & \text{if } s(x) > 0 \\ u^- & \text{if } s(x) < 0 \end{cases}$$

- Suppose at time t_s sliding surface is reached, and an ideal sliding motion takes place, i.e., $s(t) = 0\ \forall t > t_s$. This implies $\dot{s}(t) = 0,\ \forall t \geq t_s$.
- Thus, the dynamics of the sliding variable is

$$\dot{s} = \frac{\partial s}{\partial x}\dot{x} = G(x)f(x) + G(x)B(x)u_{\text{equivalent}} = 0 \tag{2}$$

where $G(x) = \dfrac{\partial s}{\partial x}$.

- Assuming the matrix GB is non-singular for any x, find the equivalent control $u_{\text{equivalent}}$ as the solution of the Eqn.(2)

$$u_{\text{equivalent}} = -(G(x)B(x))^{-1}G(x)f(x) \textit{ for } t \geq t_s.$$

- This is not the control signal which is actually applied to the plant but may be thought of as the control signal which is applied 'on average'.

Equivalent control method

Physical interpretation of the equivalent control

- For the occurrence of the ideal sliding mode, it was assumed that the control changes at a high (theoretically infinite) frequency such that the state vector is oriented precisely along the intersection of discontinuity surfaces.
- In reality, however, various imperfections make the state oscillate in some vicinity of the intersection, and control components are switched at finite frequency, alternatively taking the positive and negative values.
- These oscillations have high frequency as well as slow components.
- Almost all plants under control act as a low pass filter.
- Due to this low pass filter characteristic high frequency component is filtered out, and its motion in sliding mode is determined by the slow component.
- Practically, it is reasonable to assume that the equivalent control is close to the slow component of the real control, which can be derived by filtering out the high-frequency components using a low-pass filter.

Physical interpretation of the equivalent control

Mathematically, pass the discontinuous control signal through the low pass filter

$$\tau \dot{z} + z = u$$

to obtain the low frequency component $z(t)$.

The equivalent control is obtained as

$$\lim_{\tau \to 0} z = u_{\text{equivalent}}$$

where τ is the time constant of the low pass filter.

A brief review of Filippov theory

Convexification

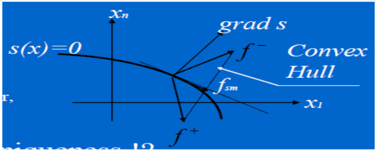

Figure: Filippov Solution

Physical interpretation of Filippov method using Regularization approach

Consider the system of the following form

$$\dot{x} = f(x, u), \quad x, f \in \mathbb{R}^n, \quad u(x) \in \mathbb{R}$$

$$u(x) = \begin{cases} u^+ & \text{if } s(x) > 0 \\ u^- & \text{if } s(x) < 0 \end{cases}$$

where the component of vector f, scalar functions u^+, u^- and $s(x)$ are continuous and smooth, and $u^+(x) \neq u^-(x)$.

A brief review of Filippov theory

Physical interpretation of Filippov method using Regularization approach

- Regularization theory studies methods for the solution to ill-posed problems (i.e., problems for which at least one of the conditions of uniqueness, the existence of the solution is not ensured)
- We assume that sliding mode occurs on the sliding surface $s(x) = 0$. Derive the motion equations using the regularization method.
- Let the discontinuous control be implemented. Control is assumed to take one of the two extreme values, u^+ or u^-, and the discontinuity points are isolated in time.
- Since discontinuity points are isolated in time, the solution exists in the conventional sense beyond the sliding surface.
- Further assume that the state velocity vector $f^+ = f_1 = f(x, u^+)$ and $f^- = f_2 = f(x, u^-)$ to be constant for some point x on the surface $s(x) = 0$ within a short interval $[t, t + \Delta t]$.

Physical interpretation of Filippov method using Regularization approach

Let the time interval Δt consists of two sets of intervals Δt_1 and Δt_2 such that $\Delta t = \Delta t_1 + \Delta t_2$, where Δt_1 and Δt_2 is the amount of that time when control of magnitude u^+ and u^- is active.

Mathematically increment of the state vector in this interval Δt is given by

$$\Delta x = f_1 \Delta t_1 + f_2 \Delta t_2$$

and the average state velocity is given as

$$\begin{aligned}\dot{x}_{\text{average}} &= \frac{\Delta x}{\Delta t} = \frac{f_1 \Delta t_1 + f_2 \Delta t_2}{\Delta t_1 + \Delta t_2} = f_1 \frac{\Delta t_1}{\Delta t_1 + \Delta t_2} + f_2 \frac{\Delta t_2}{\Delta t_1 + \Delta t_2} \\ &= \alpha f_1 + (1-\alpha) f_2\end{aligned}$$

where $\alpha = \frac{\Delta t_1}{\Delta t}$ is relative time for control to take value u^+ and $(1-\alpha)$ to take value u^- and also $0 \leq \alpha < 1$.

Physical interpretation of Filippov method using Regularization approach

To get the velocity vector $\dot{x}$ along the sliding surface we have to take limit $\Delta t \to 0$. Hence sliding motion is represented as

$$\dot{x} = \alpha f_1 + (1 - \alpha) f_2 \tag{3}$$

Remark

One can also interpret Eqn. (3) as the velocity vector in the vicinity of a point on a discontinuous surface which is complemented by a minimal convex set, and the state velocity vector of the sliding motion belongs to this set.

- Because the state trajectories during sliding mode are in the sliding surface $s = 0$, the parameter α should be selected such that the state velocity vector of the system (3) is in the tangential plane to the sliding surface.

Physical interpretation of Filippov method using Regularization approach

Mathematically, one can write

$$\dot{s} = \nabla[s(x)].\dot{x} = \nabla[s(x)](\alpha f_1 + (1-\alpha) f_2) = 0$$

where $\nabla[s(x)] = \left[\frac{\partial s}{\partial x_1} \cdots \frac{\partial s}{\partial x_n}\right]$. The solution of the above equation is given by

$$\alpha = \frac{\nabla(s).f_2}{\nabla(s).(f_2 - f_1)} \tag{4}$$

Substituting the α from Eqn.(4) to (3), one can get motion in sliding mode as

$$\dot{x} = f_{\text{sliding}} = \frac{\nabla(s).f_2}{\nabla(s).(f_2 - f_1)} f_1 - \frac{\nabla(s).f_1}{\nabla(s).(f_2 - f_1)} f_2$$

Remark

Sliding mode occurs in the surface $s(x) = 0$, therefore, the function s and $\dot{s}$ have different signs in the vicinity of the surface and

$$\dot{s}^+ = \nabla(s).f_1 < 0,$$
$$\dot{s}^- = \nabla(s).f_2 > 0$$

Check that $\dot{s} = \nabla(s) f_{\text{sliding}} = 0$ for the trajectories of the system (4) and show that they are confined to the switching surface $s(x) = 0$.

Thank You!

Lecture 11

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E1 248
Sliding mode control and its application

Outline

Sliding Mode Control

Consider a single-input system

$$\dot{x} = Ax + Bu + Bd, \quad x \in \mathbb{R}^n, \; u \in \mathbb{R}$$

and the sliding manifold

$$\mathcal{S} = \left\{ x \in \mathbb{R}^n : s = c^\top x = 0 \right\}.$$

- Since in this case only one control input appears, here sliding variable $s = c^\top x$ is a real-valued function.
- Once the sliding mode is enforced in the system, $s = 0$, and the trajectories are confined to $\mathcal{S}$.
- Switching/discontinuous control become active on $\mathcal{S}$.

Consider a multi-input system

$$\dot{x} = Ax + Bu + Bd, \quad x \in \mathbb{R}^n, \; u \in \mathbb{R}^m$$

Note that there will be m number of sliding variables, i.e., $s \in \mathbb{R}^m$. So, the sliding manifold is given as

$$\mathcal{S} = \left\{ x \in \mathbb{R}^n : s = c^\top x = 0 \right\}$$

where $c \in \mathbb{R}^{n \times m}$.

What are the challenges in MIMO?

- Since there are m sliding variables, it may be possible that sliding mode may occur in i-th component of s, where $i = 1, 2, \ldots, m$.
- Design a control which brings the sliding mode into the system.

Sliding Mode

Definition

The system is said to be in sliding mode if the system trajectories are confined to

$$\mathcal{S} = \bigcap_{i=1}^{m} \mathcal{S}_i$$

where the hyperplane $\mathcal{S}_i$ is given as

$$\mathcal{S}_i = \left\{ x \in \mathbb{R}^n : s_i = c_i^\top x = 0 \right\}.$$

Here, c_i is i-th column of c.

- Clearly, in multi-input case, the sliding mode is enforced only when all the components of s become zero or in other words, $s = 0$
- If $m = 1$, this definition reduces to the same as that defined in the single-input case
- In multi-input, it may happen sliding takes place in any i-th sliding hyperplane before sliding mode is enforced in the system

Sliding Mode Control

There are two schools of thought on the design of sliding mode control for MIMO systems.

Diagonalization Approach

- The matrix $c^\top B$ is transformed into diagonal form first.
- Here, the control is designed such that each control affects only the respective sliding variable.
- Each control element brings the sliding mode on its respective sliding hyperplane, i.e., u_i forces the system trajectory to the manifold $\mathcal{S}_i$ and so on.
- Sliding mode occurs only when trajectories are forced to all m-th sliding hyperplanes.

Unit Vector Approach

- Switching control is designed such that its sliding mode is enforced to the manifold $\mathcal{S}$. It is given as

$$u_{\text{dis}} = \frac{s}{\|s\|}$$

for any $s \in \mathbb{R}^m$. For $m = 1$, it reduces to $\frac{s}{|s|}$ which is equivalent to $\text{sign}(s)$.

- Here, control switches only when the trajectories reach the manifold $\mathcal{S}$.
- Compared to the diagonalization approach, the switching signal does not affect when the trajectory crosses any of these sliding hyperplanes.

Diagonalization Approach

- Differentiating $s = c^\top x$ gives $\dot{s} = c^\top Ax + c^\top Bu + c^\top Bd$.
- The sliding surface parameter is chosen such that $c^\top B$ is a diagonal matrix.
- Sliding mode control can be derived as $u = -(c^\top B)^{-1}\left(c^\top Ax + K\mathrm{sign}s\right)$.
- Here, K is a diagonal matrix with diagonal elements K_i for $i = 1, 2, \ldots, m$ and $\mathrm{sign}s = [\mathrm{sign}s_1 \ \cdots \ \mathrm{sign}s_m]^\top$.
- Thus, it can be easily written component-wise as

$$u_i = -(c^\top B)_i^{-1}\left((c^\top A)_i x + K_i \mathrm{sign}s_i\right), \qquad \forall i = 1, 2, \ldots, m.$$

The closed-loop system becomes

$$\dot{s}_i = -K_i \mathrm{sign}s_i + (c^\top B)_i d_i, \qquad \forall i = 1, 2, \ldots, m$$

where $K_i > \sup_{t\geq 0}|(c^\top B)_i d_i(t)|$. Hence, the sliding mode is enforced when all s_i's are zero.

Unit Vector Approach

Sliding Mode Control

Consider again the sliding dynamics

$$\dot{s} = c^\top Ax + c^\top Bu + c^\top Bd.$$

In order to ensure $s = 0$, design the control as

$$u = -(c^\top B)^{-1}\left(c^\top Ax + K\frac{s}{\|s\|}\right).$$

Here, the switching gain K is a scalar. The closed loop system stability can be obtained by substituting the control in $\dot{s}$ is given as

$$\dot{s} = -K\frac{s}{\|s\|} + c^\top Bd.$$

The convergence of the system trajectory to the sliding manifold $\mathcal{S}$ is discussed now. This is possible only when $s = 0$ in finite time.

Unit Vector Approach

Theorem:

Sliding mode occurs in the system with defined control if $K > \left\| c^\top B \right\| d_m$.

Proof

Consider a Lyapunov function $V = s^\top s$. Taking the derivation of s along the system trajectory, we obtain

$$
\begin{aligned}
\dot{V} &= \dot{s}^\top s + s^\top \dot{s} \\
&= -K \frac{s^\top s}{\|s\|} + d^\top B^\top c s - K \frac{s^\top s}{\|s\|} + s^\top c^\top B d \\
&= -2K\|s\| + 2 s^\top c^\top B d \\
&\leq -2K\|s\| + 2\|s\| \left\| c^\top B d \right\| .
\end{aligned}
$$

We have used the fact $s^\top s = \|s\|^2$ in third equality. If the d is bounded, i.e., $\|d\| < d_m$,

$$
\begin{aligned}
\dot{V} &\leq -2K\|s\| + 2\|s\| \left\| c^\top B \right\| d_m \\
&= -2 \left(K - \left\| c^\top B \right\| d_m \right) V^{1/2} \\
&\leq -2\eta V^{1/2}
\end{aligned}
$$

where $\eta > 0$.

Unit Vector Approach

Proof

Integrating both sides of the above differential inequality with the condition $V(t_1) = 0$ gives

$$\int_{V_0}^{0} V^{-1/2}\,\mathrm{d}V \leq -2\eta \int_{0}^{t_1} \mathrm{d}t \quad \Longrightarrow \quad -2V_0^{1/2} \leq -2\eta t_1.$$

Thus, the system trajectory reaches the sliding manifold in a finite time not larger than $t_1 \leq \frac{V_0^{1/2}}{\eta}$.
Hence, the unit vector approach to sliding mode control ensures the sliding mode in the system and the proof is completed.

Sliding Mode Design Using Full Order Lyapunov Matrix

System Description

We consider a system in regular form

$$\dot{x}_1 = A_{11}x_1 + A_{12}x_2$$
$$\dot{x}_2 = A_{21}x_1 + A_{22}x_2 + B_2u + B_2d.$$

Here, we consider a multi-input system, i.e., $u \in \mathbb{R}^m$. Though this design technique is explicitly stated for multi-input, it is easily extended to the single-input case with minor modifications.

Control Problems

In the following, we answer to the following questions:

- Can we design sliding variable (s) using the full order system rather than the reduced dynamics?
- Is it possible to characterize the system behaviour using full order dynamics of the system?

Sliding Mode Design Using Full Order Lyapunov Matrix

Review

Consider sliding function $s = c^\top x = c_1^\top x_1 + x_2$.

- We know that there exists a control (sliding mode) law such that $s = 0$ in finite time. It implies $x_2 = -c_1^\top x_1$.
- Substituting this in the upper part of system dynamics,

$$\dot{x}_1 = \left(A_{11} - A_{12}c_1^\top\right) x_1.$$

- Thus, the closed-loop system dynamics can be given as

$$\begin{aligned}\dot{x}_1 &= \left(A_{11} - A_{12}c_1^\top\right) x_1 \\ x_2 &= -c_1^\top x_1.\end{aligned}$$

Remarks

- Sliding function parameter c_1 is such that all the eigenvalue of the matrix $A_{11} - A_{12}c_1^\top$ have negative real part.
- The vector c_1 is designed from reduced dynamics.

Sliding Mode Design Using Full Order Lyapunov Matrix

Here, we show how the switching function parameters are designed from the full-order system.

Theorem

[a]Suppose that a matrix $Q > 0$ is arbitrarily chosen and $L \in \mathbb{R}^{m\times n}$ is given such that

$$(A - BL)^\top P + P(A - BL) + Q = 0 \tag{1}$$

for some $P > 0$. Then, the following statements hold:

- Sliding function parameter c_1 exists of the form

  $$c_1^\top = P_{22}^{-1} P_{12}^\top$$

  for a P satisfying (1) where $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^\top & P_{22} \end{bmatrix}$.
- For any $P > 0$ satisfying (1), the sliding function parameter $c_1^\top = P_{22}^{-1} P_{12}^\top$ is a stable sliding function parameter for the system.

[a] K.-S. Kim and Y. Park, "Sliding mode design via quadratic performance optimization with pole-clustering constraints", *SIAM Journal of Control and Optimization*, vol. 43, no. 2, pp. 670–684, 2004.

Sliding Mode Design Using Full Order Lyapunov Matrix

Proof of First Statement

Given the sliding function parameter, $c_1^\top$ is a stabilizing gain, and we show that this choice of $P > 0$ guarantees full order system stability. Since $c_1^\top$ is a stabilizing gain, the matrix

$$A_{11} - A_{12}c_1^\top$$

is stable. Define $0 < Q_r = [I_{n-m} \;\; -c_1]Q[I_{n-m} \;\; -c_1]^\top$. Then there exists $P_r > 0$ such that

$$(A_{11} - A_{12}c_1^\top)^\top P_r + P_r(A_{11} - A_{12}c_1^\top) + Q_r = 0.$$

Fix an arbitrary $P_{22} > 0$. Then let

$$P_{12} = c_1 P_{22}, \quad P_{11} = P_r + P_{12}^\top P_{22}^{-1} P_{12}, \quad L = [L_1 \;\; L_2]$$

where

$$L_1 = B_2^{-1}\left\{A_{21} + P_{22}^{-1}P_{12}^\top A_{11} + P_{22}^{-1}A_{12}^\top P_r\right\}$$

$$L_2 = B_2^{-1}\left\{A_{22} + P_{22}^{-1}P_{12}^\top A_{12} + \frac{\epsilon}{2}P_{22}^{-1}\right\}, \quad \epsilon > 0.$$

Sliding Mode Design Using Full Order Lyapunov Matrix

Construct $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^\top & P_{22} \end{bmatrix}$ using the pre-designed matrices. Then, with some manipulations, the following relation results

$$T\left((A - BL)^\top P + P(A - BL) + Q\right) T^\top = \begin{bmatrix} R & 0 \\ 0 & -\epsilon I_m \end{bmatrix} = 0$$

where $R = (A_{11} - A_{12}c_1^\top)^\top P_r + P_r(A_{11} - A_{12}c_1^\top) + Q_r$ and $T = \begin{bmatrix} I_{n-m} & -P_{12}P_{22}^{-1} \\ 0 & I_m \end{bmatrix}$.

So, the full order system is stable.

Proof of Second Statement

For a P satisfying (1), define $T_r = [I_{n-m} \ \ -P_{12}P_{22}^{-1}]$. Pre and post multiplying T_r and $T_r^\top$ on both sides of (1), gives

$$\left(A_{11} - A_{12}P_{22}^{-1}P_{12}^\top\right)^\top P_r + P_r\left(A_{11} - A_{12}P_{22}^{-1}P_{12}^\top\right) + T_rQT_r^\top = 0$$

where $P_r = P_{11} - P_{12}P_{22}^{-1}P_{12}^\top$ which is a positive definite matrix since $P > 0$. Thus by selecting $c_1^\top = P_{22}^{-1}P_{12}^\top$, the stability of the matrix is ensured, and sliding mode enforces stable motion in the system.

Sliding Mode Design Using Full Order Lyapunov Matrix

- From the above result, we see that the sliding function parameter can be chosen from the Lyapunov equation of the full-order system.
- Sliding mode is enforced in the system by the control law

$$u = -(c^\top B)^{-1}(c^\top Ax + K\mathrm{sign}s)$$

where K is a diagonal matrix and $\mathrm{sign}s = [\mathrm{sign}s_1 \ \cdots \ \mathrm{sign}s_m]^\top$.

Sliding Hyperplane Design Under Parametric Uncertainties

Consider an uncertain system[a]

$$\dot{x} = (A + \Delta A)x + Bu + Bd$$

where ΔA represents real parametric uncertainties of the form

$$\Delta A = MD(t)N.$$

Here, $M, N^\top \in \mathbb{R}^{n\times h}$, and $\mathbb{R}^{h\times h} \ni D(t) = \mathrm{diag}\{\delta_1 I_{r_1}, \ldots, \delta_p I_{r_p}\}$ s.t. $|\delta_i| \leq 1$ for $i = 1, 2, \ldots, p$ and $t \geq 0$. To see how to write ΔA as $\Delta A = MD(t)N$, we use rank factorization.

- Consider any matrix $S \in \mathbb{R}^{m\times n}$ of rank r. Then, it can be written as $S = S_1 S_2$ where $S_1 \in \mathbb{R}^{m\times r}$ and $S_2 \in \mathbb{R}^{r\times n}$.
- Write ΔA as $\Delta A = \sum_{i=1}^{p} \delta_i(t) E_i$ with rank of E_i as r_i.

[a] K.-S. Kim, Y. Park and S.-H. Oh, "Designing robust sliding hyperplanes for parametric uncertain systems: a Riccati approach", *Automatica,* vol. 36, no. 7, pp. 1041–1048, 2000.

Sliding Hyperplane Design Under Parametric Uncertainties

Then, from rank factorization, we express $E_i = \underline{E}_i \overline{E}_i$ where $\underline{E}_i$ and $\overline{E}_i$ are matrices of appropriate dimensions. So,

$$\begin{aligned}
\Delta A &= \underline{E}_1 \delta_1 \overline{E}_1 + \cdots + \underline{E}_p \delta_p \overline{E}_p \\
&= \begin{bmatrix} \underline{E}_1 & \cdots & \underline{E}_p \end{bmatrix} \begin{bmatrix} \delta_1 I_{r_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \delta_p I_{r_p} \end{bmatrix} \begin{bmatrix} \overline{E}_1 \\ \vdots \\ \overline{E}_p \end{bmatrix} = MDN.
\end{aligned}$$

Example: Consider ΔA given below with $|\delta_i| \leq 1$ for all $i = 1, 2, 3$ as

$$\begin{aligned}
\Delta A &= \begin{bmatrix} 0 & \delta_3 & \delta_1 \\ \delta_1 & 2\delta_1 & \delta_2 \\ 0 & \delta_2 & 0 \end{bmatrix} = \delta_1 \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \delta_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \delta_3 \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
&= \delta_1 E_1 + \delta_2 E_2 + \delta_3 E_3.
\end{aligned}$$

Example

To write $E_1 = \underline{E}_1 \overline{E}_1$, we first convert E_1 in reduced row echelon form (i.e., all leading coefficient is one in row echelon form) as

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then, $\underline{E}_1$ is obtained by removing all non-pivotal columns from E_1 and $\overline{E}_1$ is obtained by eliminating all zero rows from the reduced row echelon form of E_1. This is given as

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{E}_1 \overline{E}_1.$$

Similarly, E_2 can be written in row echelon form as

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Example

It can be verified as

$$E_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{E}_2 \overline{E}_2$$

$$E_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \underline{E}_3 \overline{E}_3.$$

So, we write ΔA as

$$\Delta A = \delta_1 \underline{E}_1 \overline{E}_1 + \delta_2 \underline{E}_2 \overline{E}_2 + \delta_3 \underline{E}_3 \overline{E}_3 = \begin{bmatrix} \underline{E}_1 & \underline{E}_2 & \underline{E}_3 \end{bmatrix} \begin{bmatrix} \delta_1 \overline{E}_1 \\ \delta_2 \overline{E}_2 \\ \delta_3 \overline{E}_3 \end{bmatrix}$$

$$= \begin{bmatrix} \underline{E}_1 & \underline{E}_2 & \underline{E}_3 \end{bmatrix} \begin{bmatrix} \delta_1 I_2 & 0 & 0 \\ 0 & \delta_2 I_2 & 0 \\ 0 & 0 & \delta_3 I_1 \end{bmatrix} \begin{bmatrix} \overline{E}_1 \\ \overline{E}_2 \\ \overline{E}_3 \end{bmatrix} = MDN.$$

Sliding Hyperplane Design Under Parametric Uncertainties

Definition

A system is said to be quadratically stable if there exists a linear state feedback control law $u = -Lx$, $P > 0$ and a scalar $\alpha > 0$ such that $V = x^\top Px$ is Lyapunov function and $\dot{V} < -\alpha\|x\|^2$.

Theorem

Let $d = 0$. The system is quadratically stable with linear state feedback control $u = -Lx$ if there exists $P > 0$, $X \in \mathrm{S}_{\mathrm{sym}}$ and $U \in \mathrm{S}_{\mathrm{skew}}$ such that

$$(A - BL)^\top P + P(A - BL) + Q + PMXM^\top P + (N + PMU^\top)X^{-1}(N + PMU^\top)^\top < 0$$

for some $Q \geq 0$ and

$$\mathrm{S}_{\mathrm{sym}} = \{X : XD(t) = D(t)X, X > 0\}$$
$$\mathrm{S}_{\mathrm{skew}} = \left\{U : UD(t) = D(t)U, U = -U^\top\right\}.$$

Sliding Hyperplane Design Under Parametric Uncertainties

Proof

Consider $V = x^\top P x$. Taking time derivative, we obtain

$$\dot{V} = x^\top\{(A - BL)^\top P + P(A - BL)\}x + x^\top\{PMDN + N^\top D^\top M^\top P\}x.$$

The second term can be written as

$$\begin{aligned} PMDN + N^\top D^\top M^\top P &= PMDH + H^\top D^\top M^\top P \\ &= (PMX^{1/2})(DX^{-1/2}H) + (DX^{-1/2}H)^\top (PMX^{1/2})^\top \\ &\leq PMXM^\top P + H^\top X^{-1} H \end{aligned}$$

where $H = N + UM^\top P$. The first equality holds since $U \in \mathrm{S}_{\text{skew}}$ and the second equality holds because $X \in \mathrm{S}_{\text{sym}}$, so $D = X^{1/2} D X^{-1/2}$. In the last step, we use the bounding inequality given as

$$HY + Y^\top H^\top \leq HH^\top + Y^\top Y.$$

So, combining all these, we write

$$\dot{V} \leq x^\top\{(A - BL)^\top P + P(A - BL) + PMXM^\top P + H^\top X^{-1} H\}x.$$

Sliding Hyperplane Design Under Parametric Uncertainties

Proof

In other words, if

$$(A - BL)^\top P + P(A - BL) + Q + PMXM^\top P + H^\top X^{-1} H < 0$$

then the system is quadratically stable for some $Q \geq 0$.

Theorem

Sliding mode exists in the system if there exists $P > 0$, $L \in \mathbb{R}^{m \times n}$, $X \in \mathrm{S}_{\text{sym}}$ and $U \in \mathrm{S}_{\text{skew}}$ such that

$$(A - BL)^\top P + P(A - BL) + Q + PMXM^\top P + (N + PMU^\top)X^{-1}(N + PMU^\top)^\top < 0 \tag{2}$$

for some $Q \geq 0$.

Proof

Pre and post multiplying both sides of (2) by T_r and $T_r^\top$ where $T_r = [I_{n-m} \quad -P_{12}P_{22}^{-1}]$. Then, after some simple calculation it gives $c_1^\top = P_{22}^{-1}P_{12}^\top$ such that reduced dynamics is stable.

Thank You!

Lecture 12 & 13

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E1 248
Sliding mode control and its application

Outline

Terminal Sliding Mode: Motivation

Classical Sliding Mode Control

- Most commonly used switching manifolds are the linear hyperplanes, which guarantee asymptotic stability of the system motion during sliding mode
- System states reach the equilibrium in infinite time

Motivation

- Need to improve the system performance during sliding mode
- System should reach equilibrium point as fast as possible
- A non-Lipschitz nonlinear manifold has faster convergence near the equilibrium point (i.e., origin)

Terminal Sliding Mode

- In TSM, a nonlinear sliding surface is proposed
- The equilibrium is a terminal attractor, i.e., the states can be reached in finite time and are stable
- The term terminal refers to the equilibrium which can be reached in finite time and is stable

Terminal Sliding Mode: Concept

Consider a second-order system

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = f(x) + g(x)u, \quad x_1, x_2 \in \mathbb{R}$$

Define the sliding variable as

$$s = x_2 + \beta x_1^{q/p}, \quad \beta > 0$$

where p and q are odd integers such that $q < p$. Differentiating s, we obtain

$$\dot{s} = \dot{x}_2 + \beta \frac{q}{p} x_1^{\frac{q}{p}-1} \dot{x}_1$$
$$= f(x) + g(x)u + \beta \frac{q}{p} x_1^{\frac{q-p}{p}} x_2.$$

Design the control as $u = -g^{-1}(x)\left(f(x) + \beta \frac{q}{p} x_1^{\frac{q-p}{p}} x_2 + Ksign(s)\right)$ where $K > 0$.

Then, simplifying further

$$\dot{s} = -Ksign(s).$$

Terminal Sliding Mode: Concept

Reduced Order Dynamics

During the sliding mode, we achieve $s = 0$, that implies

$$\dot{x}_1 = x_2 = -\beta x_1^{q/p}.$$

Now solving for time t_1 such that $x_1(t) = 0$ for all $t \geq t_1$ given any initial condition $x_1(t_0)$ at time $t = t_0$

$$\begin{aligned} t_1 &= t_0 - \frac{1}{\beta} \int_{x_1(t_0)}^{0} x_1^{-\frac{q}{p}} \mathrm{d}x_1 \\ &= t_0 + \frac{p}{\beta(p-q)} x_1^{\frac{p-q}{p}}(t_0). \end{aligned}$$

Note that $p - q$ is an even number. It implies that x_1 goes to zero in time t_1 and remains there for all time $t \geq t_1$. Since $\dot{x}_1 = 0$, then $x_2 = 0$ and thus, finite time stability is ensured.

Terminal Sliding Mode: Concept

Remarks

- The control expression contains negative exponent of x_1, so it becomes unbounded for $x_1 = 0$
- However, during sliding, the control can be bounded by selecting $\frac{q}{p} \in [0.5, 1)$. It can be shown as $u = -g^{-1}(x)\left(f(x) - \beta^2 \frac{q}{p} x_1^{2\left(\frac{q}{p} - \frac{1}{2}\right)} + K sign(s)\right)$

Terminal Sliding Manifold

Consider the terminal sliding manifold

$$s = x_2 + \beta x_1^{q/p}$$

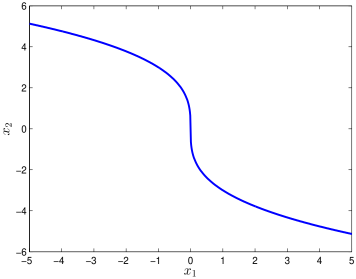


Figure: Terminal sliding manifold

Terminal Sliding Mode: Concept

Remarks

- TSM manifold is non-Lipschitz in nature
- Near origin, the convergence rate is much faster than the linear surface
- The solution of such a system reaches the equilibrium point in finite time
- If $p = q$, then TSM manifold becomes linear sliding surface

TSM for chain of integrator system with single-input

$$\begin{aligned} \dot{x}_i &= x_{i+1} \quad i = 1, 2, \ldots, n-1, \\ \dot{x}_n &= f(x) + g(x)u. \end{aligned}$$

TSM for chain of integrator system with single-input

For the n$^{\text{th}}$ order single-input system, hierarchical TSM manifolds are defined as

$$
\begin{aligned}
s_1 &= \dot{s}_0 + \beta_1 s_0^{q_1/p_1} \\
s_2 &= \dot{s}_1 + \beta_2 s_1^{q_2/p_2} \\
&\vdots \\
s_{n-2} &= \dot{s}_{n-3} + \beta_{n-2} s_{n-3}^{q_{n-2}/p_{n-2}} \\
s_{n-1} &= \dot{s}_{n-2} + \beta_{n-1} s_{n-2}^{q_{n-1}/p_{n-1}}
\end{aligned}
$$

where $s_0 = x_1$, $\beta_i > 0$, $p_i > q_i$ and p_i, q_i are positive odd integers.

The values of the integer must satisfy following for bounded control during sliding[1]

$$\frac{q_k}{p_k} > \frac{n-k}{n-k+1} \quad k = n-1, \ldots, 1.$$

[1] X. Yu and Z. Man, "Model reference adaptive control systems with terminal sliding mode", Int. J. Control, vol. 64, no. 6, pp. 1165–1176, 1996.

TSM for chain of integrator system with single-input

TSM Control

Differentiating s_{n-1}, we obtain

$$\begin{aligned}
\dot{s}_{n-1} &= \ddot{s}_{n-2} + \beta_{n-1}\frac{q_{n-1}}{p_{n-1}} s_{n-2}^{\frac{q_{n-1}}{p_{n-1}}-1} \frac{\mathrm{d}}{\mathrm{d}t} s_{n-2} \\
&= f(x) + g(x)u + \sum_{i=1}^{n-1} \frac{\mathrm{d}^i}{\mathrm{d}t^i} \beta_{n-i} s_{n-i-1}^{\frac{q_{n-i}}{p_{n-i}}}.
\end{aligned}$$

Now, design the control law

$$u = -g^{-1}(x)\left(f(x) + \sum_{i=1}^{n-1} \frac{\mathrm{d}^i}{\mathrm{d}t^i} \beta_{n-i} s_{n-i-1}^{\eta_{n-i}} + K\mathrm{sign}(s_{n-1}) \right).$$

Substituting u in the above dynamics

$$\dot{s}_{n-1} = -K\mathrm{sign}(s_{n-1}).$$

It leads to $s_{n-1} = 0$ in finite time. This implies $s_{n-2} = 0$ and subsequently $s_0 = x_1 = 0$. Thus, all the states of the system goes to zero in finite time.

Fast Terminal Sliding Mode Control

Fast Terminal Sliding Mode

Motivation

Consider the terminal sliding manifold

$$s = x_2 + \beta x_1^{q/p}.$$

During sliding $\dot{x}_1 = -\beta x_1^{\frac{q}{p}}$. It can be observed that

- if the initial condition is far away from the origin, the term $x_1^{\frac{q}{p}}$ has lesser magnitude than that of linear counterpart
- convergence can be enhanced by incorporating a linear term in the terminal sliding manifold

Fast TSM: To achieve faster convergence, a new TSM manifold is defined as

$$s = x_2 + \alpha x_1 + \beta x_1^{q/p}$$

and during sliding $\dot{x}_1 = -\alpha x_1 - \beta x_1^{\frac{q}{p}}$. Thus,

- when x_1 is far away from the origin αx_1 dominates, in other words, $\dot{x}_1 \approx -\alpha x_1$, so convergence is faster.
- for the values of x_1 close to origin, $\beta x_1^{\frac{q}{p}}$ will be the dominating term, so $\dot{x}_1 \approx -\beta x_1^{\frac{q}{p}}$.

Fast Terminal Sliding Mode

Reduced Order System

The reduced system during sliding can be given as

$$\dot{x}_1 = -\alpha x_1 - \beta x_1^{\frac{q}{p}}.$$

The time of convergence of x_1 to zero can be obtained as

$$\begin{aligned} t_1 &= t_0 + \int_{x_1(t_0)}^{0} \frac{\mathrm{d}x_1}{-\alpha x_1 - \beta x_1^{\frac{q}{p}}} \\ &= t_0 + \int_{x_1(t_0)}^{0} \frac{\mathrm{d}x_1}{-x_1^{\frac{q}{p}} \left(\alpha x_1^{1-\frac{q}{p}} + \beta\right)} \\ &= t_0 + \frac{p}{\alpha(p-q)} \left(\ln\left(\alpha x_1^{\frac{p-q}{p}}(t_0) + \beta\right) - \ln(\beta) \right) \end{aligned}$$

where t_0 is the time taken by the system to reach the fast terminal sliding manifold.

Fast Terminal Sliding Mode

Single-input System

$$\dot{x}_i = x_{i+1} \quad i = 1, 2, \ldots, n-1,$$
$$\dot{x}_n = f(x) + g(x)u.$$

For this n^{th} order single-input system, hierarchical TSM manifolds are defined as

$$\begin{aligned}
s_1 &= \dot{s}_0 + \alpha_1 s_0 + \beta_1 s_0^{q_1/p_1} \\
s_2 &= \dot{s}_1 + \alpha_2 s_1 + \beta_2 s_1^{q_2/p_2} \\
&\vdots \\
s_{n-2} &= \dot{s}_{n-3} + \alpha_{n-2} s_{n-3} + \beta_{n-2} s_{n-3}^{q_{n-2}/p_{n-2}} \\
s_{n-1} &= \dot{s}_{n-2} + \alpha_{n-1} s_{n-2} + \beta_{n-1} s_{n-2}^{q_{n-1}/p_{n-1}}
\end{aligned}$$

where $s_0 = x_1$, $\beta_i > 0$, $p_i > q_i$ and p_i, q_i are positive odd integers. The values of integer must satisfy for bounded control during sliding given as[a]

$$\frac{q_k}{p_k} > \frac{n-k-1}{n-k} \quad k = n-1, \ldots, 1.$$

[a] X. Yu and Z. Man, "Fast terminal sliding-mode control design for nonlinear dynamical systems", IEEE Trans. Circuit Systems-1, Fundam. Theory and Appl., vol. 49, no. 2, pp. 261–264, 2002.

Non-Singular Terminal Sliding Mode Control

Non-Singular Terminal Sliding Mode

Terminal Sliding Mode Control

Recall the TSM control law

$$u = -g^{-1}(x)\left(f(x) + \beta\frac{q}{p}x_1^{\frac{q}{p}-1}x_2 + K sign(s)\right).$$

We see that the exponent of x_1 is $\frac{q}{p} - 1 < 0$. So, when system trajectories cross $x_1 = 0$ axis, then the control law becomes infinite. Such a controller can not be applied to the system and it is called singularity in the TSM.

Non-Singular Terminal Sliding Mode

To avoid such a situation, a new terminal manifold is proposed called non-singular terminal sliding mode (NTSM)[a]

$$s = x_1 + \frac{1}{\beta^{\frac{p}{q}}}x_2^{\frac{p}{q}}, \quad 1 < \frac{p}{q} < 2$$

The TSM and NTSM surfaces are equivalent to each other when $s = 0$.

[a] Y. Feng, X. Yu and Z. Man, "Non-singular terminal sliding mode control of rigid manipulators", Automatica, vol. 38, no. 12, pp. 2159–2167, 2002.

Non-Singular Terminal Sliding Mode

Equivalence between TSM and NTSM

- It is to be noted that $x_1^{\frac{q}{p}}$ is an odd function, i.e., $(-x_1)^{\frac{q}{p}} = -x_1^{\frac{q}{p}}$.
- One way to realize this, we can take $x_1^{\frac{q}{p}} = |x_1|^{\frac{q}{p}}\mathrm{sign}(x_1)$.

Now, we shall see the equivalence between TSM and NTSM when $s = 0$. From NTSM with $s = 0$, we have

$$x_1 = -\frac{1}{\beta^{\frac{p}{q}}}|x_2|^{\frac{p}{q}}\mathrm{sign}(x_2).$$

From this, we conclude that $\mathrm{sign}(x_1) = -\mathrm{sign}(x_2)$. Multiplying both sides by $\mathrm{sign}(x_1)$ and then taking $\frac{q}{p}$ power on both sides (use the fact $|x_1| = x_1\mathrm{sign}(x_1)$)

$$\beta|x_1|^{\frac{q}{p}} = |x_2|.$$

Multiplying both sides by $\mathrm{sign}(x_2)$, it yields

$$-\beta|x_1|^{\frac{q}{p}}\mathrm{sign}(x_1) = x_2.$$

This, in other words, equal to $x_2 = -\beta x_1^{\frac{q}{p}}$. Thus, the time taken by the system to reach $x_1 = 0$ is the same as that of TSM.

Non Singular Terminal Sliding Mode

Finite time reachability to NTSM manifold

Differentiating s

$$\begin{aligned}\dot{s} &= \dot{x}_1 + \frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}\dot{x}_2 \\ &= x_2 + \frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}(f(x)+g(x)u).\end{aligned}$$

Design the control law as given below

$$u = -g^{-1}(x)\left(f(x) + \beta^{\frac{p}{q}}\frac{q}{p}x_2^{2-\frac{p}{q}} + K\mathrm{sign}(s)\right).$$

Substituting for u in $\dot{s}$, we obtain

$$\dot{s} = -\frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}K\mathrm{sign}(s).$$

Non Singular Terminal Sliding Mode

Finite time reachability to NTSM manifold

- To show convergence to origin, we consider $V = \frac{1}{2}s^2$. Differentiating V along the system trajectories

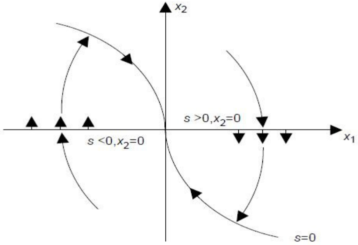
$$\dot{V} = s\dot{s} = -s\left(\frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}K\text{sign}(s)\right)$$

  which, on further simplification

$$\dot{V} = -\frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}K|s|.$$

- Define $\rho(x_2) := \frac{1}{\beta^{\frac{p}{q}}}\frac{p}{q}x_2^{\frac{p}{q}-1}K$.
- Then $s\dot{s} = -\rho(x_2)|s|$. If $x_2 \neq 0$, we have $\rho(x_2) > 0$.
- That means the trajectories are attracted towards the NTSM manifold, and hence, finite time convergence is achieved.
- For $x_2 = 0$, we write $\dot{x}_2 = -K\text{sign}(s)$.

Non Singular Terminal Sliding Mode



Finite time reachability to NTSM manifold

- If $s > 0$, then $\dot{x}_2 = -K$. Similarly for $s < 0$, we have $\dot{x}_2 = K$.
- It implies that there exists a small vicinity $|x_2| < \delta$ around $x_2 = 0$ such that for $s > 0$, we have $\dot{x}_2 = -K$. Similarly for $s < 0$.
- Then x_2 decreases for $s > 0$ and increases for $s < 0$. So, the sliding trajectories will cross the boundaries $x_2 = \delta$ and $x_2 = -\delta$ in finite time and similarly for $s < 0$.
- Therefore, the trajectories are attracted towards the NTSM manifold in finite time. Thus, the proof is completed.

Prescribed Convergence Law

Prescribed Convergence Law

Consider a second order system

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = f(x) + g(x)u$$

where $g(x) \neq 0$. Define the sliding variable as $s = x_2 + \beta|x_1|^{\frac{1}{2}}\mathrm{sign}(x_1)$. The control law is given as

$$u = -g^{-1}(x)(f(x) + \alpha\mathrm{sign}(s))$$

where $\alpha > \frac{\beta^2}{2}$. Substituting the control in the system dynamics, we obtain

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\alpha\mathrm{sign}(s).$$

It can be seen that trajectories are driven by a constant rate gain, hence the name prescribed convergence law[a][b].

[a] A. Levant, "Universal single-input-single-output(SISO) sliding mode controllers with finite time convergence", IEEE Trans. Autom. Control, vol. 46, no. 9, pp. 1447–1451, 2001.

[b] A. Levant, "Higher-order sliding modes, differentiation and output-feedback control", Int. J. Control, vol. 76, no. 9/10, pp. 924–941, 2003.

Prescribed Convergence Law: Proof

Differentiating s and substituting for u

$$\dot{s} = -\alpha \mathrm{sign}(s) + \frac{1}{2}\beta |x_1|^{-\frac{1}{2}} x_2.$$

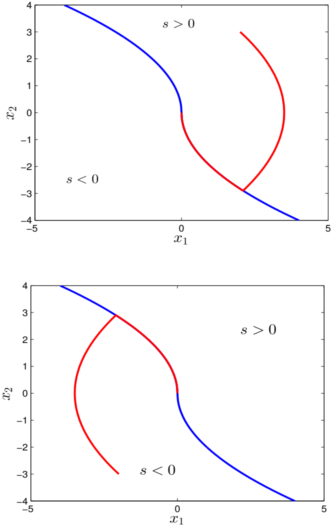
It can be noted that the initial conditions may be located either in $s > 0$ or $s < 0$ ($s = 0$ is trivial). Consider $s > 0$ and then

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -\alpha.$$

Due to geometric reasons, the system trajectories decrease and eventually hit the curve $s = 0$ on the way. Similarly, for the case $s < 0$ as the dynamics takes the form

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = \alpha.$$

It implies that the trajectories increase and hit $s = 0$ in finite time.



Prescribed Convergence Law: Proof

When $s = 0$, we have $x_2 = -\beta|x_1|^{\frac{1}{2}}\mathrm{sign}(x_1)$, so

$$\begin{aligned}\dot{s} &= -\alpha\mathrm{sign}(s) - \frac{1}{2}\beta^2\mathrm{sign}(x_1)\\ &\leq -\eta\mathrm{sign}(s).\end{aligned}$$

Thus, once the trajectories hit $s = 0$, it can never leave it provided $\alpha > \frac{\beta^2}{2}$, and hence, the sliding mode is enforced in finite time.

System Dynamics

During sliding, we obtain $\dot{x}_1 = -\beta|x_1|^{\frac{1}{2}}\mathrm{sign}(x_1)$. Consider $V = \frac{1}{2}x_1^2$. Then,

$$\begin{aligned}\dot{V} &= x_1\dot{x}_1 = -x_1\beta|x_1|^{\frac{1}{2}}\mathrm{sign}(x_1) = -\beta|x_1|^{\frac{3}{2}}\\ &= -\beta 2^{\frac{3}{4}}V^{\frac{3}{4}}.\end{aligned}$$

We see that V goes to zero in time $t_1 = t_0 + \frac{4}{\beta 2^{3/4}}V^{\frac{1}{4}}(t_0)$. Thus, finite time stability is ensured.

Summary

Remarks

- Prescribed convergence law and TSM are similar except in their control structures.
- NTSM is proposed to avoid the singularity issue in TSM.
- There is no singularity in the prescribed convergence law.
- These control structures belong to second-order sliding mode control.
- We will study second-order sliding mode control in the future.

Thank You!

Lecture 14

Sliding Mode Observer

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Sliding mode control and its application

Outline

Sliding Mode Observer: Utkin's Observer (1981)

Consider a linear time-invariant system

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

where $u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$ and $p \geq m$.

Sliding mode observer is structurally similar except in their innovative correction term. At first, the system is transformed to

$$\dot{x}_1 = A_{11}x_1 + A_{12}y + B_1u$$
$$\dot{y} = A_{21}x_1 + A_{22}y + B_2u.$$

Remark

There always exist a transformation such that $x \mapsto Tx =: (x_1, y)$ and this transformation is given as

$$T = \begin{bmatrix} N_c^\top \\ C \end{bmatrix}$$

where $N_c \in \mathbb{R}^{n\times(n-p)}$ spans the null space of C. The new output matrix is $CT^{-1} = [0 \quad I_p]$.

Sliding Mode Observer

The sliding mode observer is proposed as

$$\dot{\hat{x}}_1 = A_{11}\hat{x}_1 + A_{12}\hat{y} + B_1 u - L_1 L_2 \text{sign}(y - \hat{y})$$
$$\dot{\hat{y}} = A_{21}\hat{x}_1 + A_{22}\hat{y} + B_2 u + L_2 \text{sign}(y - \hat{y})$$

where gain L_2 is chosen such that sliding mode occurs in finite time on the manifold

$$\mathcal{S} = \{(e_1, e_y) : e_y = 0\}.$$

Here $e_1 = x_1 - \hat{x}_1$ and $e_y = y - \hat{y}$.

Observer Error Dynamics

Using the error variables, the error dynamics can be given as

$$\dot{e}_1 = A_{11}e_1 + A_{12}e_y + L_1 L_2 \text{sign}(e_y)$$
$$\dot{e}_y = A_{21}e_1 + A_{22}e_y - L_2 \text{sign}(e_y).$$

Now selecting the gain L_2 such that $L_2 > A_{21}e_1 + A_{22}e_y$, we obtain the error dynamics as

$$\dot{e}_y \leq -\eta \text{sign}(e_y).$$

Sliding Mode Observer

Observer Error Dynamics

Thus, $e_y^\top \dot{e}_y \leq -\eta\|e_y\|$ and achieve $\hat{y} = y$ in finite time. When $e_y = 0$, equivalent control is

$$L_2 \mathrm{sign}(e_y)\big|_{\mathrm{eq}} = A_{21} e_1.$$

Then, reduced dynamics is given as

$$\dot{e}_1 = A_{11} e_1 + L_1 A_{21} e_1 = (A_{11} + L_1 A_{21}) e_1.$$

Design L_1 such that $A_{11} + L_1 A_{21}$ is Hurwitz and e_1 goes to zero asymptotically.

Remark

It is to be noted that for the sliding mode to occur, i.e., $e_y = 0$, the $(e_1, e_y) \in \Omega$, where

$$\Omega = \{(e_1, e_y) : \|A_{21}\|\|e_1\| + \lambda_{\max}\{A_{22}\}\|e_y\| < L_2 - \eta\}.$$

for any $\eta < L_2$.

Sliding Mode Observer

Another Approach

- Consider the observer error dynamics

$$\dot{e}_1 = A_{11}e_1 + A_{12}e_y + L_1L_2\mathrm{sign}(e_y)$$
$$\dot{e}_y = A_{21}e_1 + A_{22}e_y - L_2\mathrm{sign}(e_y).$$

- Define transformation $(e_1, e_y) \mapsto T(e_1, e_y) =: (\tilde{e}_1, \tilde{e}_y) = (\tilde{e}_1, e_y)$ where

$$T = \begin{bmatrix} I_{n-p} & L_1 \\ 0 & I_p \end{bmatrix}.$$

- Using this transformation, we obtain

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y$$
$$\dot{e}_y = A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - L_2\mathrm{sign}(e_y).$$

where $\tilde{A}_{11} = A_{11} + L_1A_{21}$, $\tilde{A}_{12} = \tilde{A}_{11}L_1 + A_{12} + L_1A_{22}$ and $\tilde{A}_{22} = A_{21}L_1 + A_{22}$. Define the set

$$\widetilde{\Omega} = \left\{(\tilde{e}_1, e_y) : \|A_{21}\|\|\tilde{e}_1\| + \|\tilde{A}_{22}\|\|e_y\| < L_2 - \eta\right\}.$$

Sliding Mode Observer

- Now design $L_2 > A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y$ such that $(\tilde{e}_1, e_y) \in \widetilde{\Omega}$. Then,

$$\dot{e}_y \leq -\eta \mathrm{sign}(e_y)$$

and finally $e_y^\top \dot{e}_y \leq -\eta \|e_y\|$. Thus, $e_y = 0$ in finite time.

- The reduced order dynamics is given as

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1.$$

- Since (A, C) is observable pair, it implies that (A_{11}, A_{21}) is also observable pair. Then, L_1 can be chosen such that $\tilde{A}_{11}$ is Hurwitz and $\tilde{e}_1$ goes to zero asymptotically.
- Eventually, it makes e_1 to go to zero asymptotically.[1][2]

[1] S. Drakunov and V. Utkin, "Sliding mode observer: tutorial", *In the Proc. 34th Conf. Dec. Control*, New Orleans, LA, USA, Dec, 1995.

[2] [Chap. 6]C. Edwards and S. K. Spurgeon, *Sliding Mode Control: Theory and Application*, CRC Press, Taylor and Francis Group, New York, 1998.

Sliding Mode Observer

Remarks

- In the following dynamics

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y$$
$$\dot{e}_y = A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - L_2\text{sign}(e_y)$$

  the variable $\tilde{e}_1$ converges to zero asymptotically only after the sliding mode is enforced, i.e., $e_y = 0$. So, the trajectory $\tilde{e}_1$ may not decrease until the sliding mode starts.

- A modification to this observer is proposed by Edwards and Spurgeon[a] to overcome this problem. It is shown that with this modified observer, the estimation error converges to zero even before sliding mode takes place.

[a] C. Edwards and S. Spurgeon, "On the development of discontinuous observers", *International Journal of Control*, vol. 59, no. 5, pp. 1211–1229, 1994.

Sliding Mode Observer

Modified Observer

The modified observer is given with linear injection terms as

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1 + \tilde{A}_{12}e_y - G_1 e_y$$
$$\dot{e}_y = A_{21}\tilde{e}_1 + \tilde{A}_{22}e_y - G_2 e_y - L_2\mathrm{sign}(e_y).$$

Here, the gains are selected as $G_1 = \tilde{A}_{12}$ and $G_2 = \tilde{A}_{22} - A^s_{22}$, where A^s_{22} is any stable design matrix of appropriate dimension. Using this, the above dynamics reduces to

$$\dot{\tilde{e}}_1 = \tilde{A}_{11}\tilde{e}_1$$
$$\dot{e}_y = A_{21}\tilde{e}_1 + A^s_{22}e_y - L_2\mathrm{sign}(e_y).$$

- The error system is asymptotically stable for $L_2 \equiv 0$ because the poles of the combined system are given by $\lambda\{\tilde{A}_{11}\}\bigcup\lambda\{A^s_{22}\}$ and so lie in the open left half complex plane.
- In the original Utkin observer, the switching action was potentially required to make the error system stable.
- The addition of a Luenberger type gain matrix, feeding back the output error, yields the potential to provide robustness against certain classes of uncertainty by virtue of the discontinuous component $L_2\mathrm{sign}(e_y)$.

Sliding Mode Observers for Linear Uncertain Systems

Synthesis of a Discontinuous Observer

Consider the dynamical system

$$\dot{x}(t) = Ax(t) + Bu(t) + D\xi(t, x, u)$$
$$y(t) = Cx(t)$$

where $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}, C \in \mathbb{R}^{p\times n}$ and $D \in \mathbb{R}^{n\times q}$ where $p \geq q$. The function $\xi : \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^q$ is unknown but bounded so that

$$\|\xi(t, x, u)\| \leq r_1 \|u\| + \alpha(t, y)$$

where r_1 is a known scalar and $\alpha : \mathbb{R}_+ \times \mathbb{R}^p \to \mathbb{R}_+$ is a known function.

The system can be written as

$$\dot{x}_1(t) = A_{11}x_1(t) + A_{12}y(t) + B_1 u(t)$$
$$\dot{y}(t) = A_{21}x_1(t) + A_{22}y(t) + B_2 u(t) + D_2\xi.$$

where $x_1 \in \mathbb{R}^{(n-p)}, y \in \mathbb{R}^P$ and the matrix A_{11} has stable eigenvalues.

Sliding Mode Observers for Linear Uncertain Systems

A canonical Form for Observer Design

Consider the observer of the form

$$\dot{\hat{x}}_1(t) = A_{11}\hat{x}_1(t) + A_{12}\hat{y}(t) + B_1 u(t) - A_{12} e_y(t)$$
$$\dot{\hat{y}}(t) = A_{21}\hat{x}_1(t) + A_{22}\hat{y}(t) + B_2 u(t) - (A_{22} - A_{22}^s) e_y(t) + \nu$$

where A_{22}^s is a stable design matrix and $e_y = \hat{y} - y$. Let $P_2 \in \mathbb{R}^{p\times p}$ be symmetric positive definite Lyapunov matrix for A_{22}^s.

Define discontinuous vector ν as

$$\nu = \begin{cases} -\rho(t,y,u)\,\|D_2\|\,\frac{P_2 e_y}{\|P_2 e_y\|}, & \text{if } e_y \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

where the scalar function $\rho : \mathbb{R}_+ \times \mathbb{R}^p \times \mathbb{R}^m \to \mathbb{R}_+$ satisfies

$$\rho(t,x,u) \geq r_1\,\|u\| + \alpha(t,y) + \gamma_0$$

γ_0 is a positive scalar.

Sliding Mode Observers for Linear Uncertain Systems

If state estimation error $e_1 = \hat{x}_1 - x_1$, then

$$\begin{aligned}\dot{e}_1(t) &= A_{11} e_1(t) \\ \dot{e}_y(t) &= A_{21} e_1(t) + A_{22}^s e_y(t) + \nu - D_2 \xi\end{aligned}$$

Proposition

There exists a family of symmetric positive definite matrices P_2 such that the uncertain dynamical error system above is quadratically stable.

Proof

- Let $Q_1 \in \mathbb{R}^{(n-p)\times(n-p)}$ and $Q_2 \in \mathbb{R}^{p\times p}$ be symmetric positive definite design matrices.
- Define $P_2 \in \mathbb{R}^{p\times p}$ to be the unique symmetric positive definite solution to the Lyapunov equation

$$P_2 A_{22}^s + \left(A_{22}^s\right)^T P_2 = -Q_2$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

Using the computed value of P_2 define

$$\hat{Q} = A_{21}^T P_2 Q_2^{-1} P_2 A_{21} + Q_1$$

and $\hat{Q} = \hat{Q}^T > 0$.

Let $P_1 \in \mathbb{R}^{(n-p)\times(n-p)}$ be the unique symmetric positive definite solution to the Lyapunov equation

$$P_1 A_{11} + A_{11}^T P_1 = -\hat{Q}$$

Consider the candidate Lyapunov function

$$V(e_1, e_y) = e_1^T P_1 e_1 + e_y^T P_2 e_y$$

Taking time derivative of the Lyapunov function along the system trajectory

$$\dot{V} = -e_1^T \hat{Q} e_1 + e_1^T A_{21}^T P_2 e_y + e_y^T P_2 A_{21} e_1 - e_y^T Q_2 e_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

It is easy to verify that

$$\left(e_y - Q_2^{-1}P_2A_{21}e_1\right)^T Q_2 \left(e_y - Q_2^{-1}P_2A_{21}e_1\right)$$
$$= e_y^T Q_2 e_y - e_1^T A_{21}^T P_2 e_y - e_y^T P_2 A_{21} e_1 + e_1^T A_{21}^T P_2 Q_2^{-1} P_2 A_{21} e_1.$$

Denote $\widetilde{e}_y = e_y - Q_2^{-1}P_2A_{21}e_1$. Then,

$$\begin{aligned}\dot{V} &= -e_1^T \hat{Q} e_1 + e_1^T A_{21}^T P_2 Q_2^{-1} P_2 A_{21} e_1 - \widetilde{e}_y^T Q_2 \widetilde{e}_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi \\ &= -e_1^T Q_1 e_1 - \widetilde{e}_y^T Q_2 \widetilde{e}_y + 2e_y^T P_2 \nu - 2e_y^T P_2 D_2 \xi \\ &= -e_1^T Q_1 e_1 - \widetilde{e}_y^T Q_2 \widetilde{e}_y - 2\rho(t,x,u)\left\|D_2\right\|\left\|P_2 e_y\right\| - 2e_y^T P_2 D_2 \xi.\end{aligned}$$

In the last equality, we use $e_y^T P P e_y = (e_y^T P)(e_y^T P)^T = \|e_y^T P\|^2$.

Recalling the fact that $\rho(t,x,u) \geq r_1\|u\| + \alpha(t,y) + \gamma_0$, above equation becomes

$$\begin{aligned}\dot{V} &\leq -e_1^T Q_1 e_1 - \widetilde{e}_y^T Q_2 \widetilde{e}_y - 2\rho(t,x,u)\left\|D_2\right\|\left\|P_2 e_y\right\| + 2\left\|D_2\right\|\left(r_1\|u\| + \alpha(y)\right)\left\|P_2 e_y\right\| \\ &\leq -e_1^T Q_1 e_1 - \widetilde{e}_y^T Q_2 \widetilde{e}_y - 2\gamma_0\left\|D_2\right\|\left\|P_2 e_y\right\| < 0.\end{aligned}$$

Hence, the error system is quadratically stable.

Sliding Mode Observers for Linear Uncertain Systems

Sliding Motion

Consider the hyperplane in the error space given by

$$S_0 = \{e \in \mathbb{R}^n : Ce = 0\}$$

Corollary

An ideal sliding motion takes place on S_0 with the above designed ν.

Proof

Consider the Lyapunov function $V_s(e_y) = e_y^T P_2 e_y$.

Differentiating

$$\begin{aligned} \dot{V}_s &= -e_y^T Q_2 e_y + 2e_y^T P_2 A_{21} e_1 + 2e_y^T P_2(\nu - D_2\xi) \\ &\leq 2\left\|P_2 e_y\right\| \left\|A_{21} e_1\right\| - 2\gamma_0 \left\|D_2\right\| \left\|P_2 e_y\right\|. \end{aligned}$$

Sliding Mode Observers for Linear Uncertain Systems

Proof

Now, we see that

$$\|P_2 e_y\|^2 = \left(P_2^{1/2} e_y\right)^T P_2 \left(P_2^{1/2} e_y\right) \geq \lambda_{\min}(P_2) \left\|P_2^{1/2} e_y\right\|^2 = \lambda_{\min}(P_2) V_s.$$

The first equality follows from $\|P_2 e_y\|^2 = \langle P_2 e_y, P_2 e_y \rangle = \langle P_2^{1/2} e_y, (P_2^{1/2})^\top P_2 e_y \rangle = \langle P_2^{1/2} e_y, P_2 P_2^{1/2} e_y \rangle = (P_2^{1/2} e_y)^T P_2 (P_2^{1/2} e_y)$.

Hence in the domain

$$\Omega = \{(e_1, e_y) : \|A_{21} e_1\| < \|D_2\| \gamma_0 - \eta\}$$

where η is a small positive integer. It follows that

$$\dot{V}_s < -2\eta \|P_2 e_y\| \leq -2\eta \sqrt{\lambda_{\min}(P_2)} \sqrt{V_s}.$$

Hence output error e_y enters Ω in finite time and remains there.

Thank You!